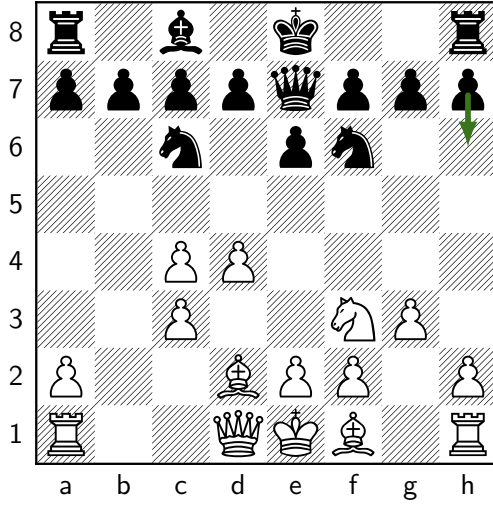


Melanie-ANALOGY — Bogo-Indian

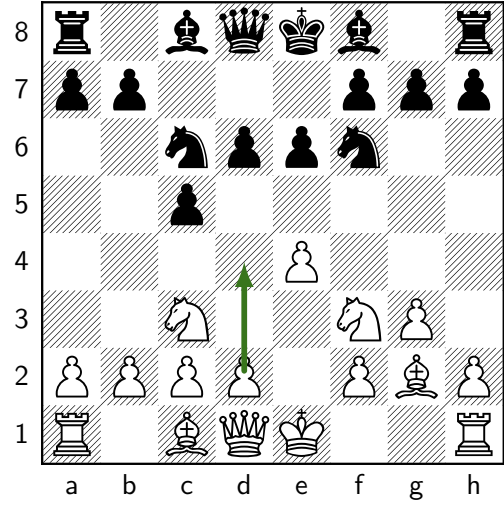
Melanie (Deep_Vision)

Abstract

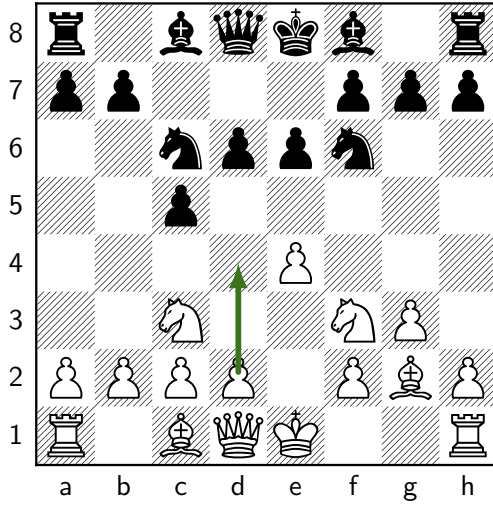
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie's move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen's role) are called out, as no material/eval engine explains a move that way.



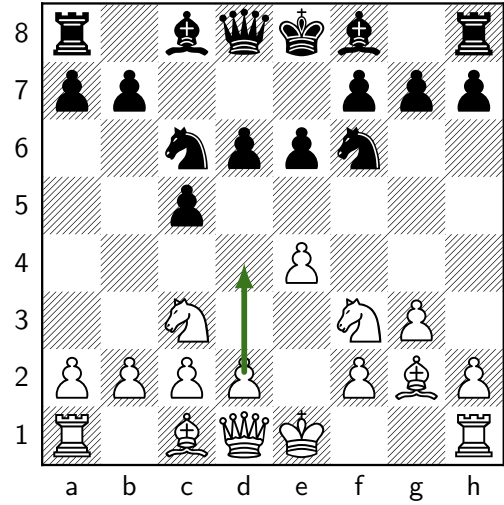
(a)



(b)

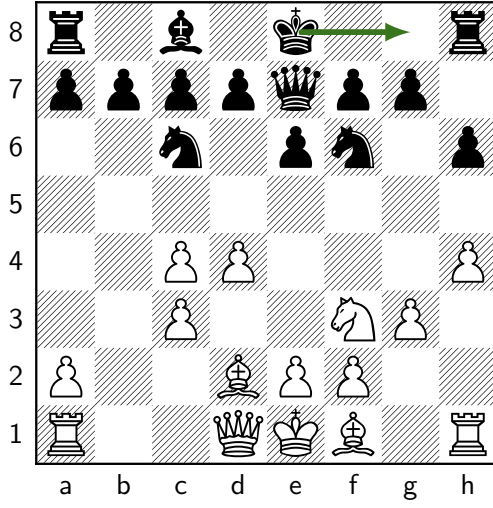


(c)

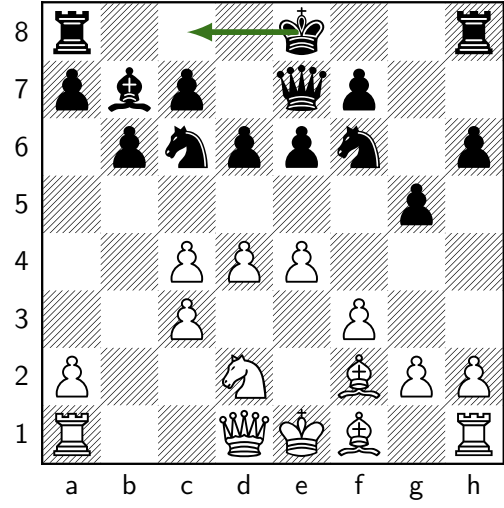


(d)

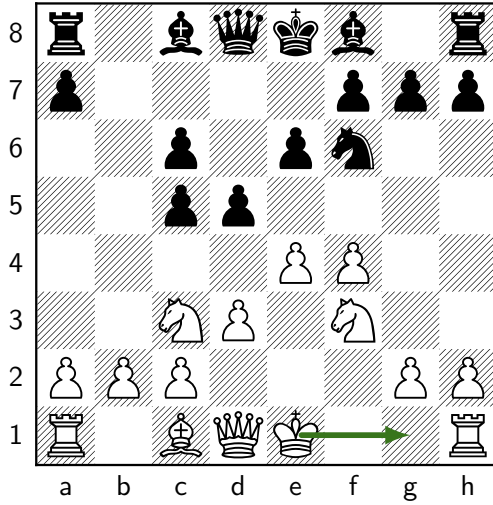
Figure 1: **Move 7, Black: h6.** (a) Our position; Melanie plays h6 (h7h6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2061` (ply 10) [structural distance $d=62$, QAP obj=23]: the master played d4; under μ our $Ph2$ plays the role of that game’s $Pd2$; cross-type roles: our $Rh8$ as their $Qd1$, our $pc4$ as their $Rh1$; transferred move h7h6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2062` (ply 10) [structural distance $d=62$, QAP obj=23]: the master played d4; under μ our $Ph2$ plays the role of that game’s $Pd2$; cross-type roles: our $Rh8$ as their $Qd1$, our $pc4$ as their $Rh1$; transferred move h7h6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1629` (ply 10) [structural distance $d=62$, QAP obj=23]: the master played d4; under μ our $Ph2$ plays the role of that game’s $Pd2$; cross-type roles: our $Rh8$ as their $Qd1$, our $pc4$ as their $Rh1$; transferred move h7h6 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to h6 (summed QAP objective 231, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



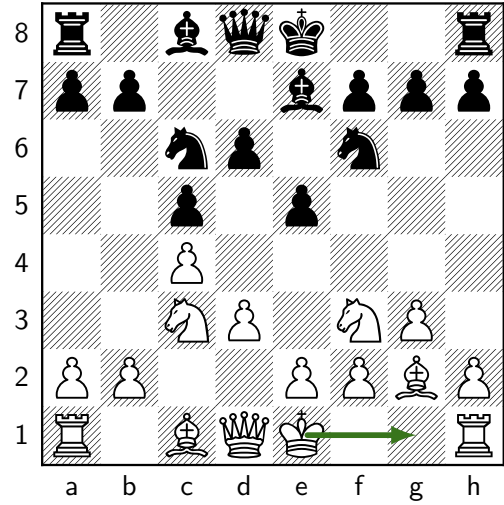
(a)



(b)

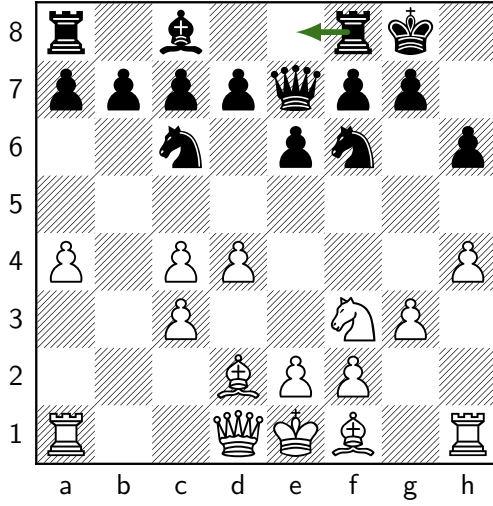


(c)

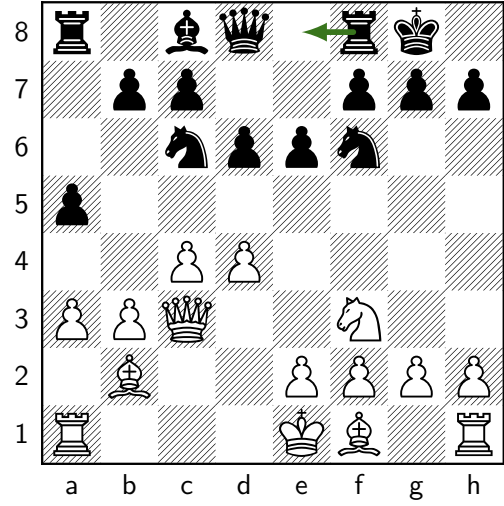


(d)

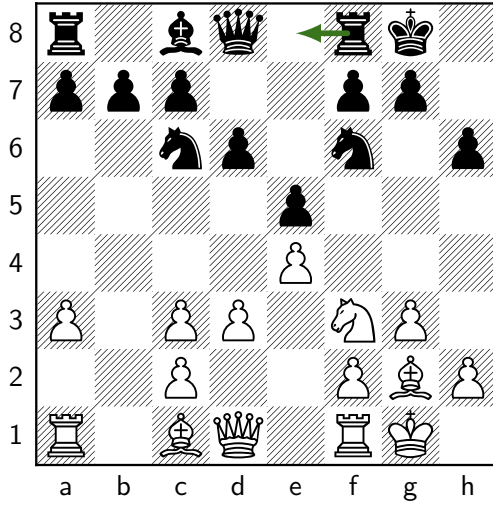
Figure 2: **Move 8, Black: 0-0.** (a) Our position; Melanie plays 0-0 (e8g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1864` (ply 23) [structural distance $d=74$, QAP obj=40]: the master played 0-0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Qe7* as their *Ra8*, our *Bc8* as their *Rh8*; transferred move e8g8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#2427` (ply 14) [structural distance $d=71$, QAP obj=27]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *qd1* as their *Ra1*, our *rh1* as their *Ph2*; transferred move e8g8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#794` (ply 12) [structural distance $d=66$, QAP obj=26]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *pc3* as their *Ra1*, our *nf3* as their *Bc1*; transferred move e8g8 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to 0-0 (summed QAP objective 200, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



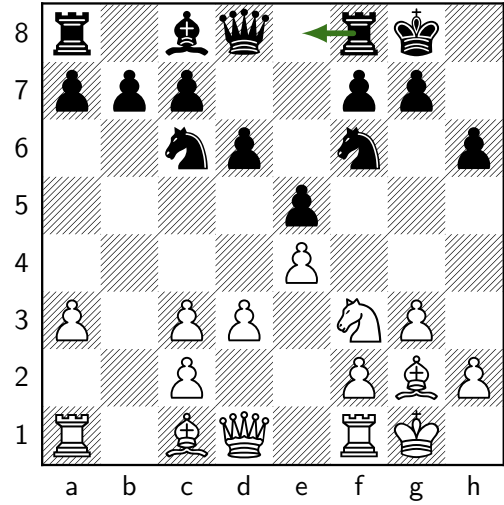
(a)



(b)

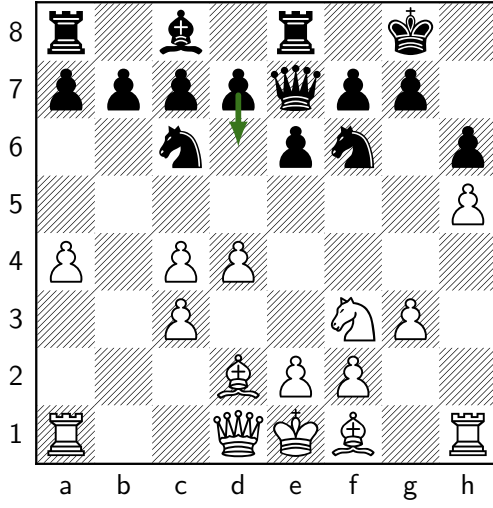


(c)

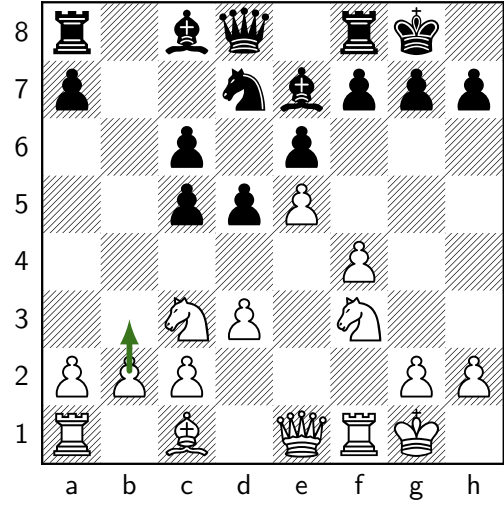


(d)

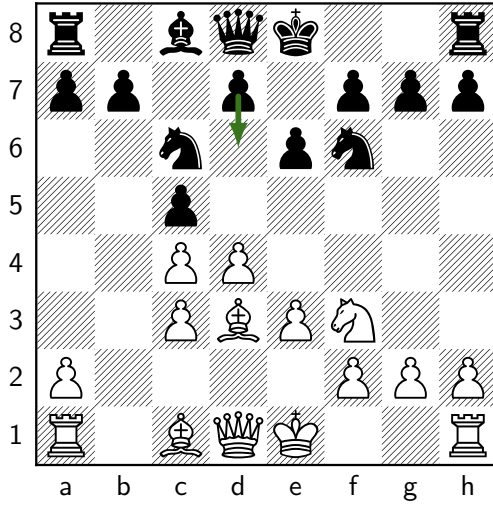
Figure 3: **Move 9, Black: Re8.** (a) Our position; Melanie plays Re8 (f8e8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#856` (ply 17) [structural distance $d=61$, QAP obj=40]: the master played Re8; under μ our *Rf1* plays the role of that game’s *Rf1*; cross-type roles: our *Nf6* as their *Ra8*, our *ra1* as their *Qd8*; transferred move f8e8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3679` (ply 17) [structural distance $d=66$, QAP obj=39]: the master played Re8; under μ our *Rf1* plays the role of that game’s *Rf1*; cross-type roles: our *pc4* as their *Ra8*, our *Nf6* as their *Bc8*; transferred move f8e8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3680` (ply 17) [structural distance $d=66$, QAP obj=39]: the master played Re8; under μ our *Rf1* plays the role of that game’s *Rf1*; cross-type roles: our *pc4* as their *Ra8*, our *Nf6* as their *Bc8*; transferred move f8e8 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Re8 (summed QAP objective 213, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



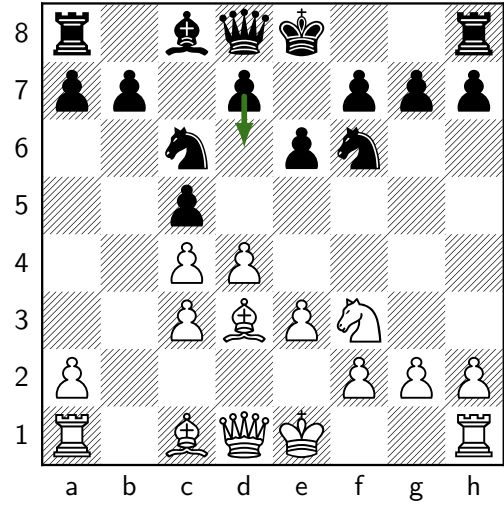
(a)



(b)

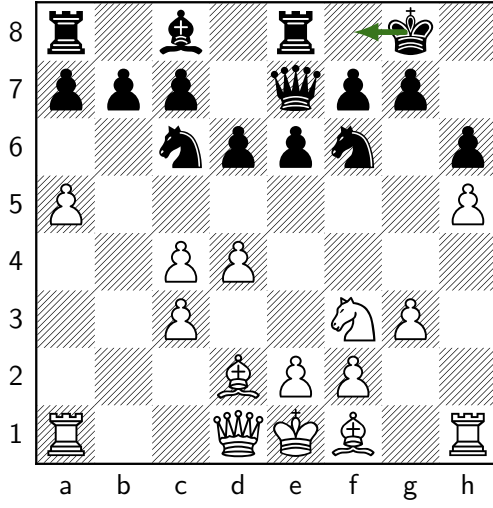


(c)

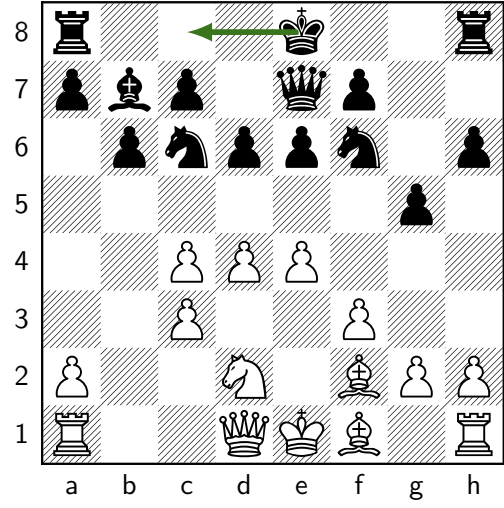


(d)

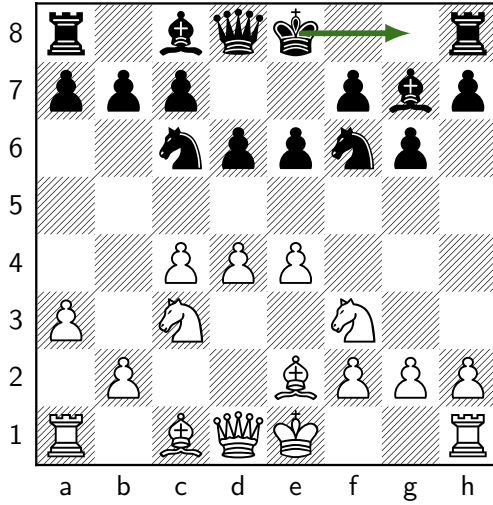
Figure 4: **Move 10, Black: d6.** (a) Our position; Melanie plays d6 (d7d6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2427` (ply 20) [structural distance $d=76$, QAP obj=44]: the master played b3; under μ our $Pd2$ plays the role of that game’s $Pb2$; cross-type roles: our $Pb7$ as their $Ra1$, our $Re8$ as their $Bc1$; transferred move d7d6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1891` (ply 13) [structural distance $d=71$, QAP obj=34]: the master played d6; under μ our $Pd2$ plays the role of that game’s $Pd2$; cross-type roles: our $Re8$ as their $Nf6$, our $Nc6$ as their $Pc5$; transferred move d7d6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1892` (ply 13) [structural distance $d=71$, QAP obj=34]: the master played d6; under μ our $Pd2$ plays the role of that game’s $Pd2$; cross-type roles: our $Re8$ as their $Nf6$, our $Nc6$ as their $Pc5$; transferred move d7d6 (**endorses this move**). Synthesis: 12 retrieved analogs transferred to d6 (summed QAP objective 419, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



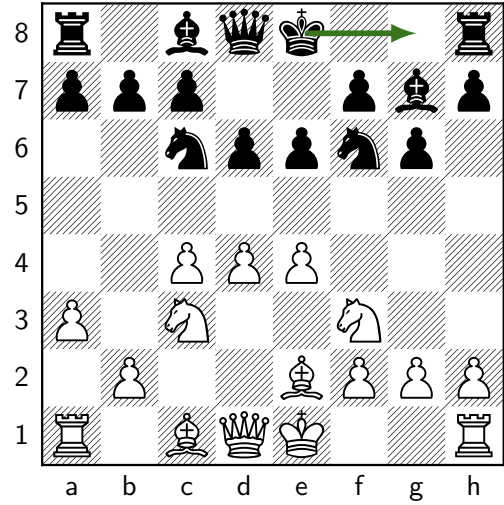
(a)



(b)

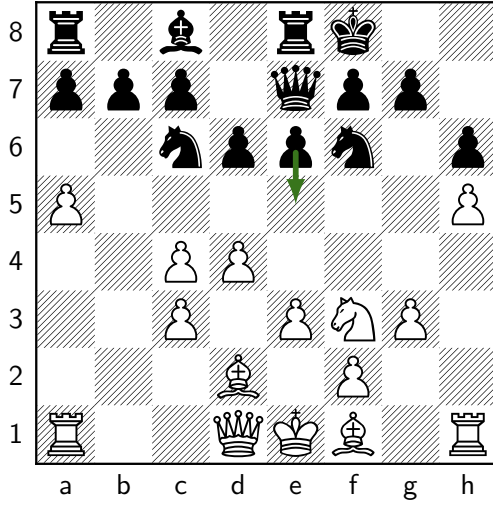


(c)

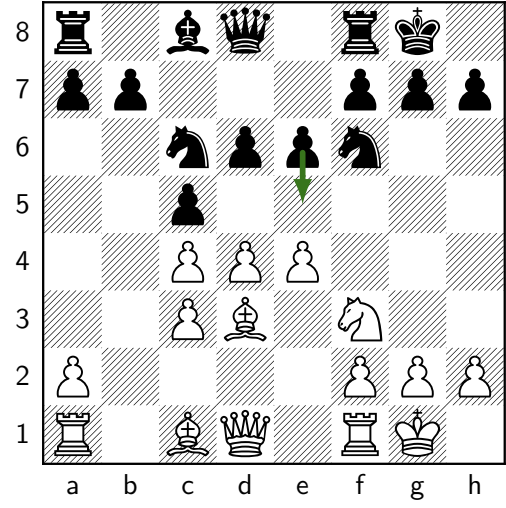


(d)

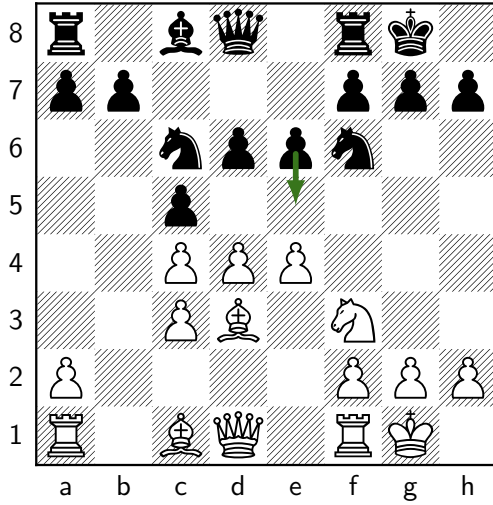
Figure 5: **Move 11, Black: Kf8.** (a) Our position; Melanie plays Kf8 (g8f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1864` (ply 23) [structural distance $d=67$, QAP obj=49]: the master played 0-0-0; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Pa7* as their *Rh8*, our *Pf7* as their *Bb7*; transferred move g8f8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#600` (ply 13) [structural distance $d=86$, QAP obj=44]: the master played 0-0; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *qd1* as their *Ra8*, our *Pf7* as their *Qd8*; transferred move g8f8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#601` (ply 13) [structural distance $d=86$, QAP obj=44]: the master played 0-0; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *qd1* as their *Ra8*, our *Pf7* as their *Qd8*; transferred move g8f8 (**endorses this move**). Synthesis: 11 retrieved analogs transferred to Kf8 (summed QAP objective 464, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



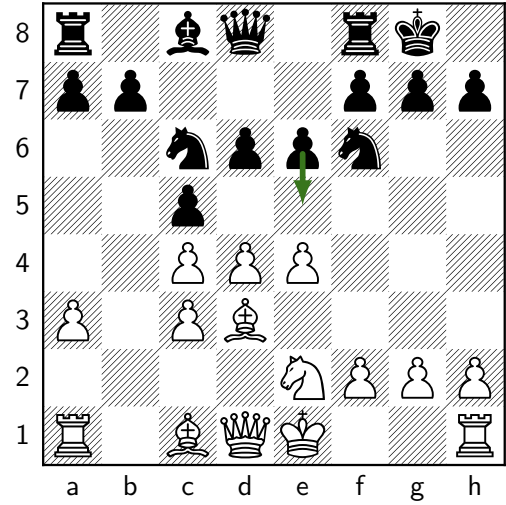
(a)



(b)

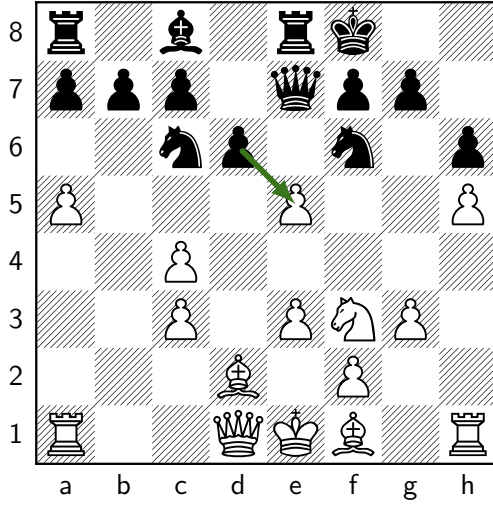


(c)

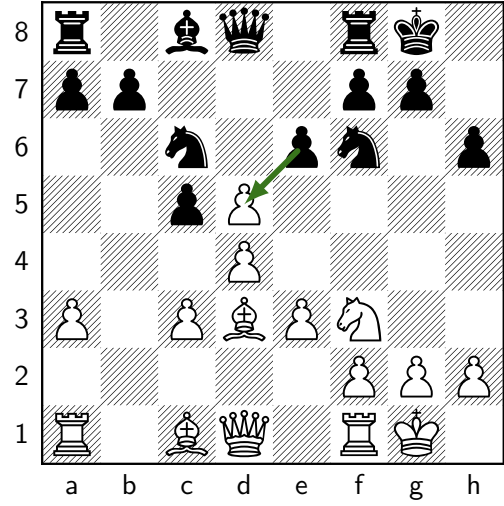


(d)

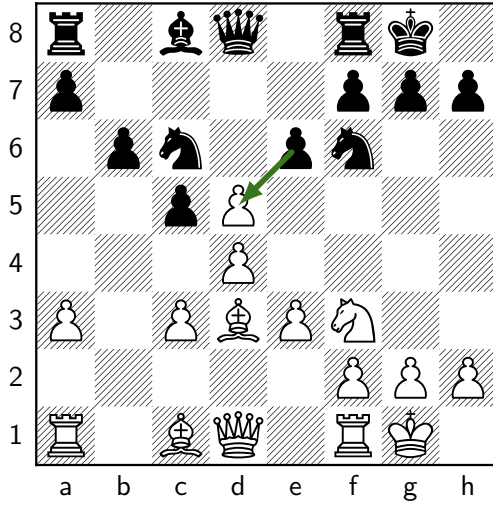
Figure 6: **Move 12, Black: e5.** (a) Our position; Melanie plays e5 (e6e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#1891` (ply 17) [structural distance $d=73$, QAP obj=47]: the master played e5; under μ our $Pe3$ plays the role of that game’s $Pe3$; cross-type roles: our $Pf7$ as their $Bc8$, our $Nc6$ as their $Pc5$; transferred move e6e5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#1892` (ply 17) [structural distance $d=73$, QAP obj=47]: the master played e5; under μ our $Pe3$ plays the role of that game’s $Pe3$; cross-type roles: our $Pf7$ as their $Bc8$, our $Nc6$ as their $Pc5$; transferred move e6e5 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#476` (ply 17) [structural distance $d=69$, QAP obj=43]: the master played e5; under μ our $Pe3$ plays the role of that game’s $Pe3$; cross-type roles: our $Pd6$ as their $Ra8$, our $Nc6$ as their $Pc5$; transferred move e6e5 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to e5 (summed QAP objective 260, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



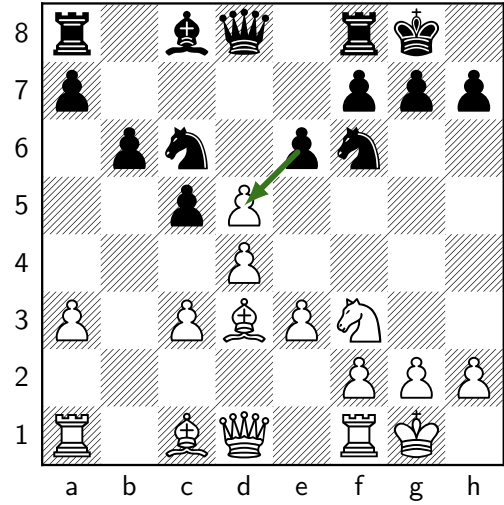
(a)



(b)

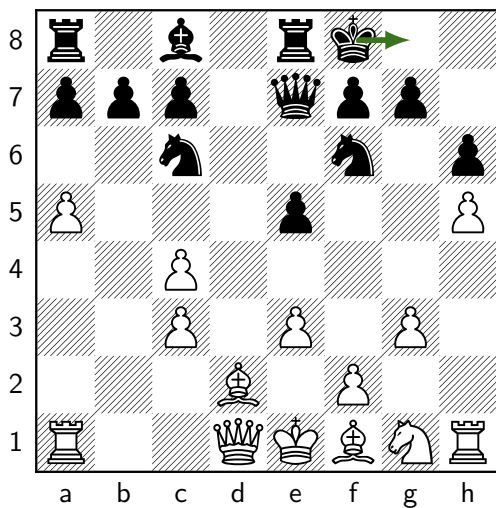


(c)

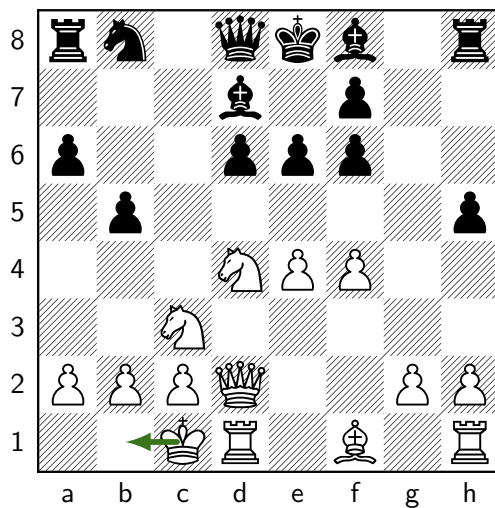


(d)

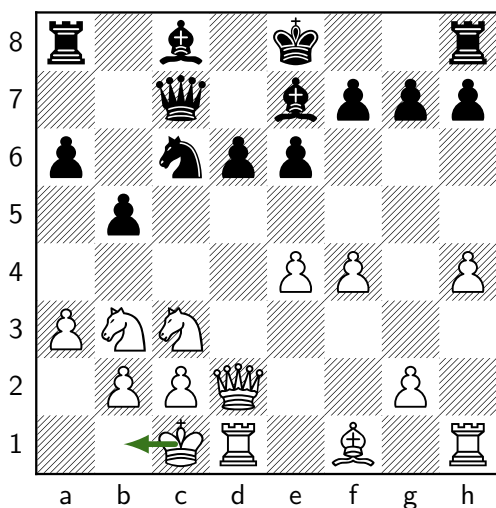
Figure 7: **Move 13, Black: dxe5.** (a) Our position; Melanie plays dxe5 (d6e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `cctl_top_p0000#1263` (ply 19) [structural distance $d=87$, QAP obj=88]: the master played `exd5`; under μ our `Pd3` plays the role of that game’s `Pe3`; cross-type roles: our `pa5` as their `Ra8`, our `Pc7` as their `Bc8`; transferred move `d6e5` (**endorses this move**). (c) Retrieved from `cctl_top_p0002#1691` (ply 19) [structural distance $d=88$, QAP obj=87]: the master played `exd5`; under μ our `Pd3` plays the role of that game’s `Pe3`; cross-type roles: our `Pb7` as their `Bc8`, our `Ph6` as their `bd3`; transferred move `d6e5` (**endorses this move**). (d) Retrieved from `cctl_top_p0003#2438` (ply 19) [structural distance $d=88$, QAP obj=87]: the master played `exd5`; under μ our `Pd3` plays the role of that game’s `Pe3`; cross-type roles: our `Pb7` as their `Bc8`, our `Ph6` as their `bd3`; transferred move `d6e5` (**endorses this move**). Synthesis: 13 retrieved analogs transferred to `dxe5` (summed QAP objective 739, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



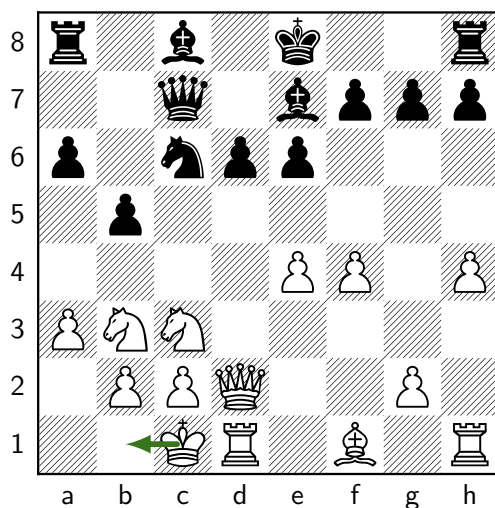
(a)



(b)

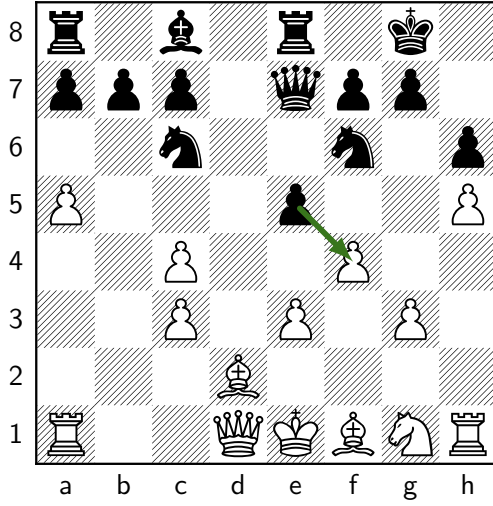


(c)

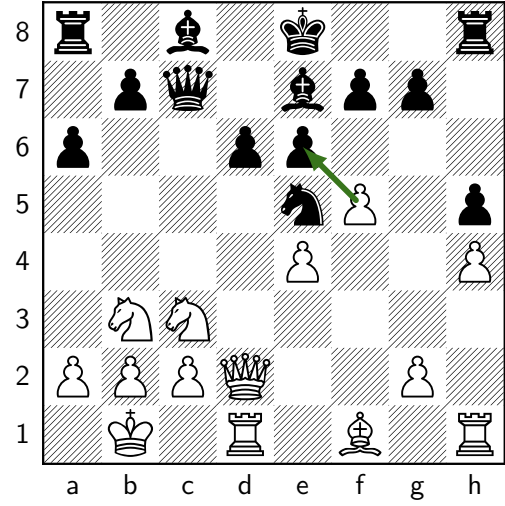


(d)

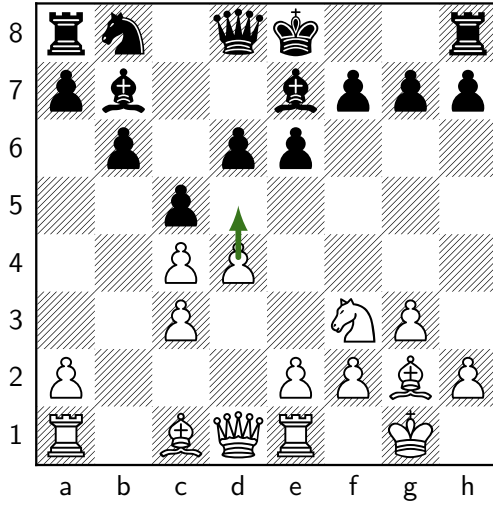
Figure 8: **Move 14, Black: Kg8.** (a) Our position; Melanie plays Kg8 (f8g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3100` (ply 22) [structural distance $d=63$, QAP obj=67]: the master played Kb1; under μ our Kf1 plays the role of that game’s Kc1; cross-type roles: our Nf6 as their Bf1, our Qe7 as their Rh1; transferred move f8g8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3906` (ply 26) [structural distance $d=62$, QAP obj=45]: the master played Kb1; under μ our Kf1 plays the role of that game’s Kc1; cross-type roles: our Qe7 as their Rd1, our Nc6 as their Bf1; transferred move f8g8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1926` (ply 26) [structural distance $d=62$, QAP obj=45]: the master played Kb1; under μ our Kf1 plays the role of that game’s Kc1; cross-type roles: our Qe7 as their Rd1, our Nc6 as their Bf1; transferred move f8g8 (**endorses this move**). Synthesis: 14 retrieved analogs transferred to Kg8 (summed QAP objective 484, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



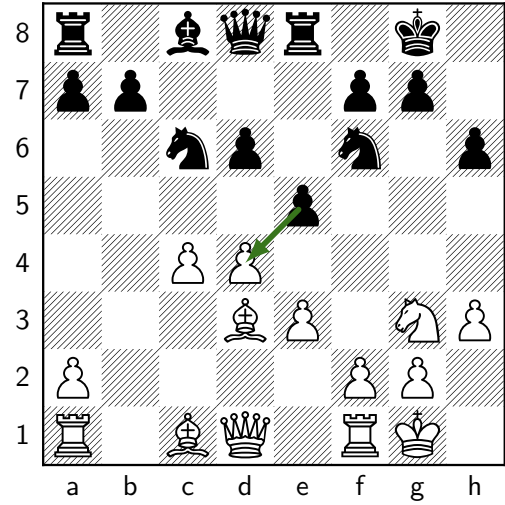
(a)



(b)

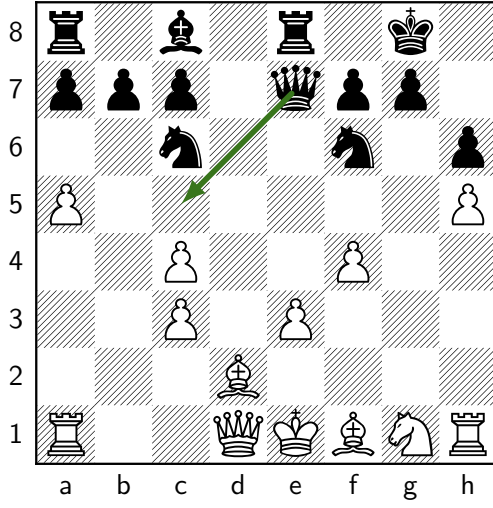


(c)

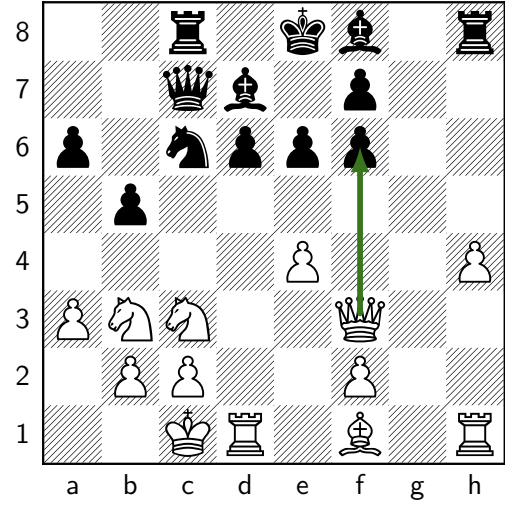


(d)

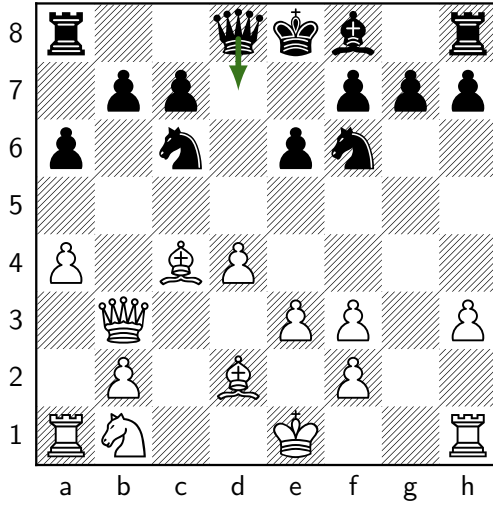
Figure 9: **Move 15, Black: exf4.** (a) Our position; Melanie plays **exf4** (e5f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1679` (ply 28) [structural distance $d=71$, QAP obj=56]: the master played **fxe6**; under μ our *Pe4* plays the role of that game’s *Pf5*; cross-type roles: our *Bc8* as their *Rd1*, our *Ra8* as their *Bf1*; transferred move **e5f4** (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3845` (ply 18) [structural distance $d=73$, QAP obj=53]: the master played **d5**; under μ our *Pe4* plays the role of that game’s *Pd4*; cross-type roles: our *pg3* as their *Ra1*, our *qd1* as their *Bc1*; transferred move **e5f4** (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2203` (ply 25) [structural distance $d=72$, QAP obj=52]: the master played **exd4**; under μ our *Pe4* plays the role of that game’s *Pe4*; cross-type roles: our *qd1* as their *Ra8*, our *Nc6* as their *Re8*; transferred move **e5f4** (**endorses this move**). Synthesis: 5 retrieved analogs transferred to **exf4** (summed QAP objective 261, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



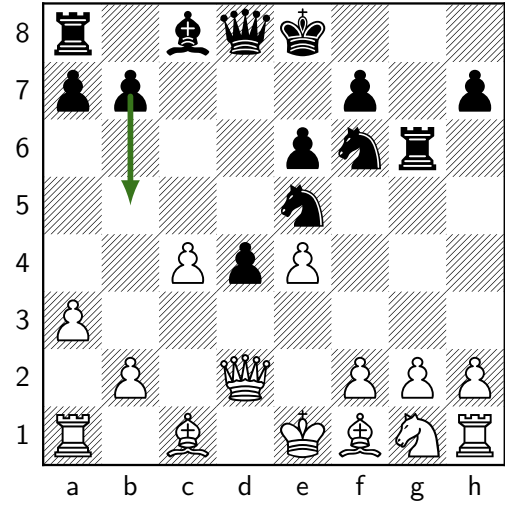
(a)



(b)

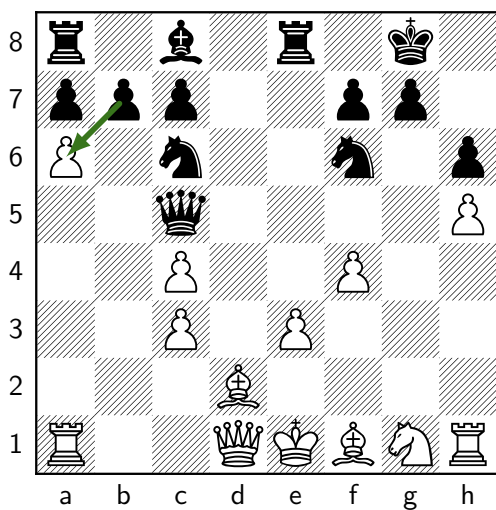


(c)

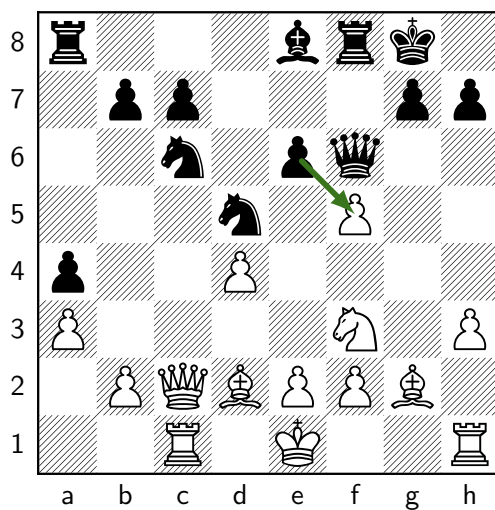


(d)

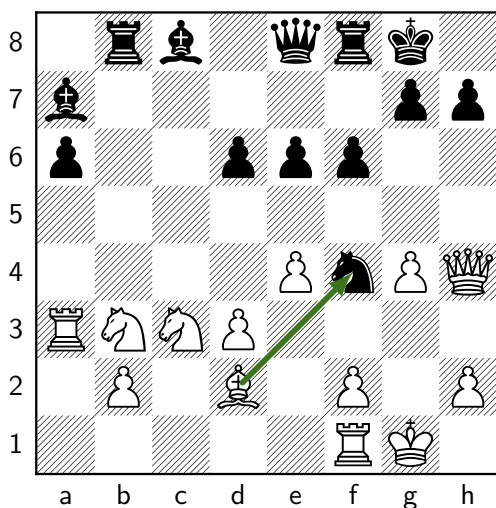
Figure 10: **Move 16, Black: Qc5.** (a) Our position; Melanie plays Qc5 (e7c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#2357` (ply 28) [structural distance $d=66$, QAP obj=91]: the master played Qxf6; under μ our Nc3 plays the role of that game’s Qf3 (a **cross-type** correspondence); cross-type roles: our Qe7 as their Rd1, our Nf6 as their Bf1; transferred move c6a5. (c) Retrieved from `ccrl_top_p0001#1262` (ply 19) [structural distance $d=71$, QAP obj=83]: the master played Qd7; under μ our Qe2 plays the role of that game’s Qd1; cross-type roles: our pc4 as their Bf8, our Ph6 as their Rh8; transferred move e7f8. (d) Retrieved from `ccrl_top_p0002#2045` (ply 27) [structural distance $d=57$, QAP obj=73]: the master played b5; under μ our Pb2 plays the role of that game’s Pb2; cross-type roles: our Bc8 as their Ra8, our Re8 as their Bc8; transferred move b7b5. Synthesis: no retrieved analog transferred to Qc5 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



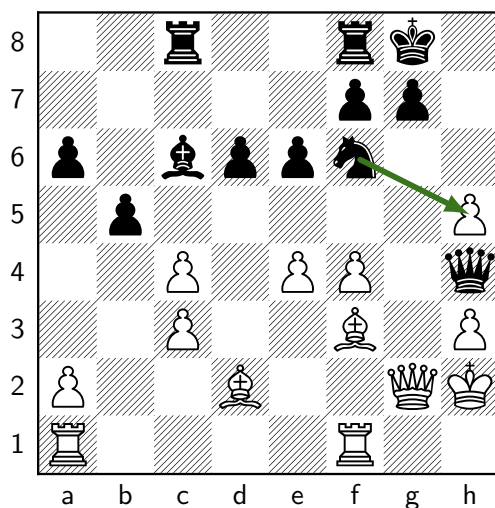
(a)



(b)

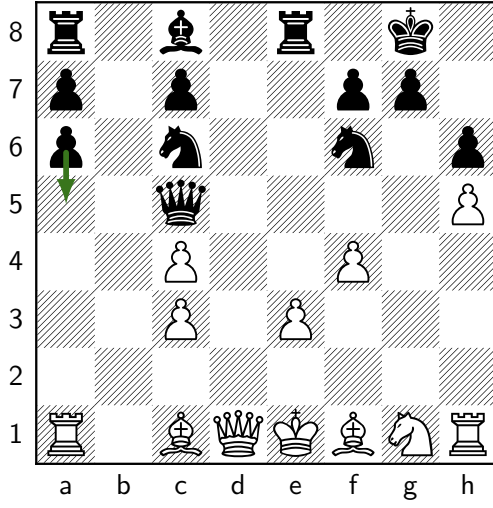


(c)

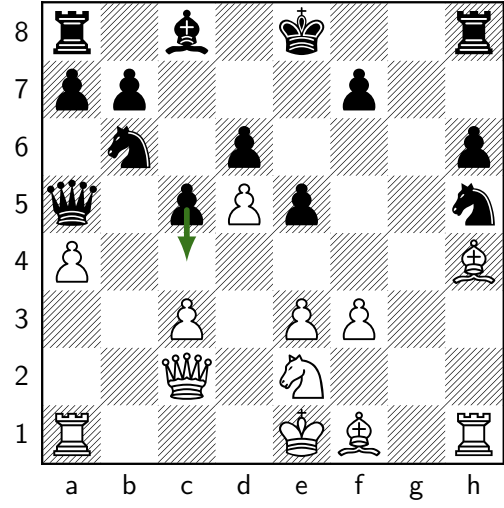


(d)

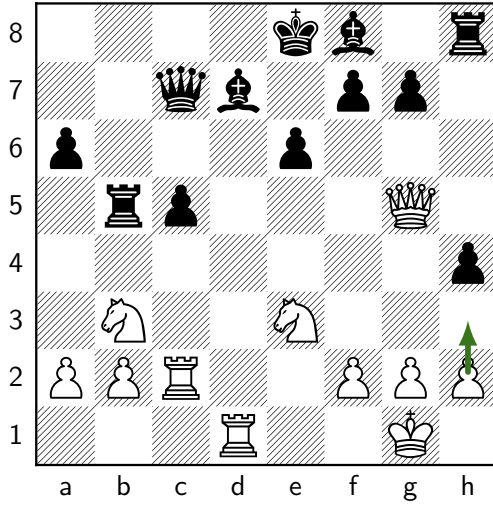
Figure 11: **Move 17, Black: bxa6.** (a) Our position; Melanie plays bxa6 (b7a6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3977` (ply 35) [structural distance $d=108$, QAP obj=84]: the master played **exf5**; under μ our *Pb2* plays the role of that game’s *Pe3*; cross-type roles: our *Nf6* as their *Rf8*, our *bd2* as their *Pb7*; transferred move **b7a6** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2607` (ply 38) [structural distance $d=110$, QAP obj=46]: the master played **Bxf4**; under μ our *Pb2* plays the role of that game’s *Bd2* (a **cross-type** correspondence); cross-type roles: our *Nf6* as their *Rf1*, our *Ra8* as their *Pb2*; transferred move **b7a6** (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2797` (ply 43) [structural distance $d=109$, QAP obj=153]: the master played **Nxh5**; under μ our *Qc4* plays the role of that game’s *Nf3* (a **cross-type** correspondence); cross-type roles: our *Nf6* as their *Rc8*, our *Bc8* as their *Pa6*; transferred move **c5a5**. Synthesis: 2 retrieved analogs transferred to **bxa6** (summed QAP objective 129, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



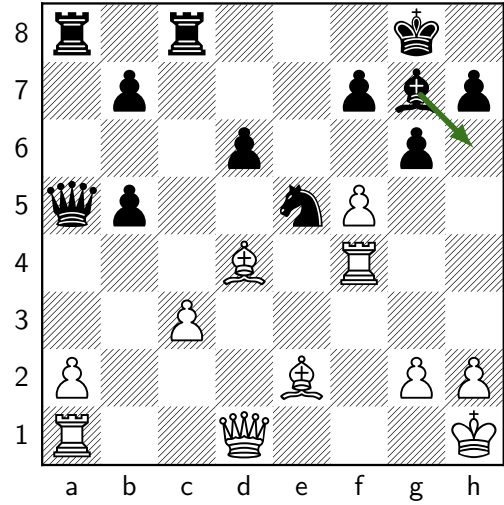
(a)



(b)

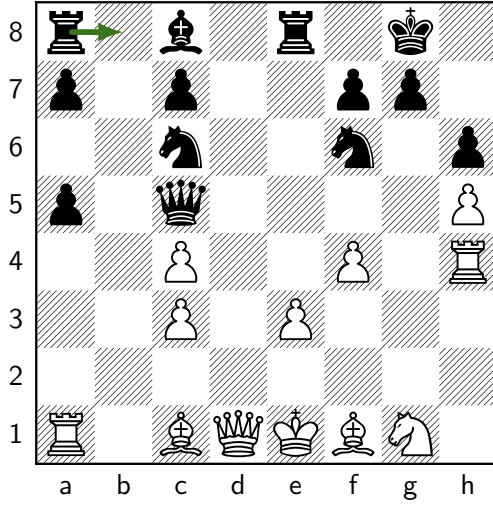


(c)

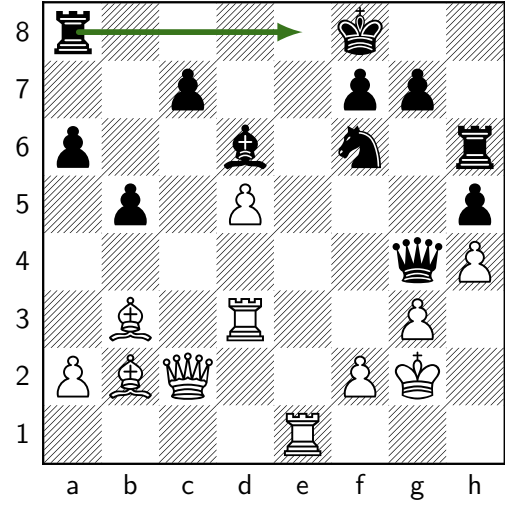


(d)

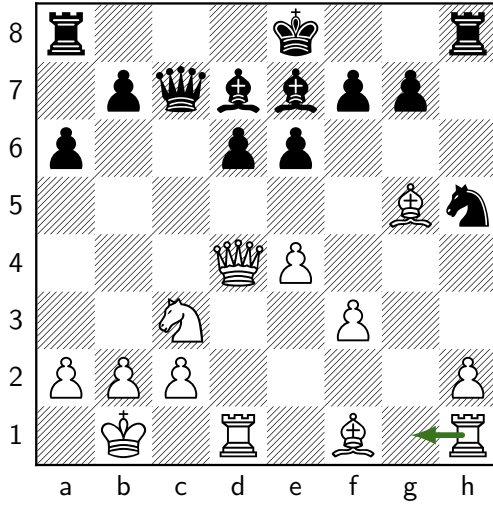
Figure 12: **Move 18, Black: a5**. (a) Our position; Melanie plays a5 (a6a5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#553` (ply 35) [structural distance $d=73$, QAP obj=79]: the master played c4; under μ our $Pa3$ plays the role of that game’s $Pc4$; cross-type roles: our $Pa7$ as their $Ra8$, our $Nc6$ as their $Bc8$; transferred move a6a5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#722` (ply 38) [structural distance $d=101$, QAP obj=71]: the master played h3; under μ our $Pa3$ plays the role of that game’s $Ph2$; cross-type roles: our $Nf6$ as their $Rd1$, our $ra1$ as their $Pa2$; transferred move a6a5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1319` (ply 41) [structural distance $d=106$, QAP obj=34]: the master played Bh6; under μ our $Pa3$ plays the role of that game’s $Bg2$ (a **cross-type** correspondence); cross-type roles: our $Nf6$ as their $Ra8$, our $Ra8$ as their $Pb7$; transferred move a6a5 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to a5 (summed QAP objective 209, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



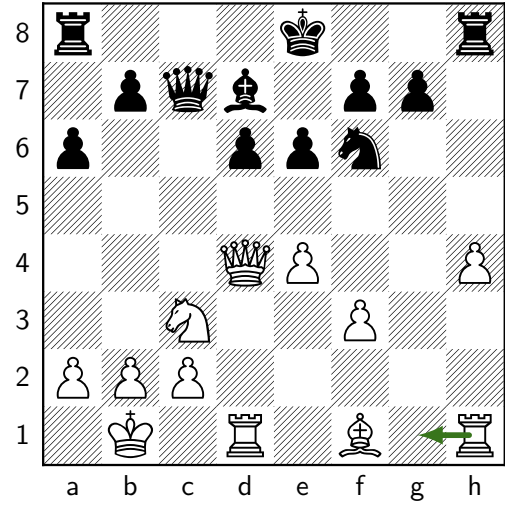
(a)



(b)

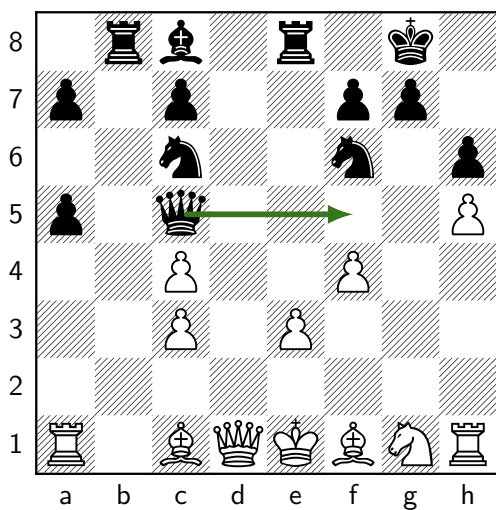


(c)

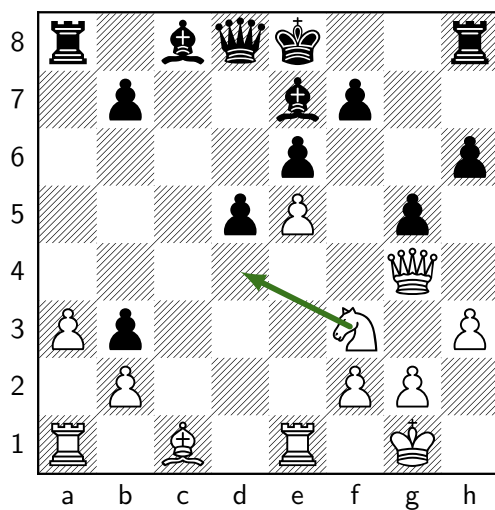


(d)

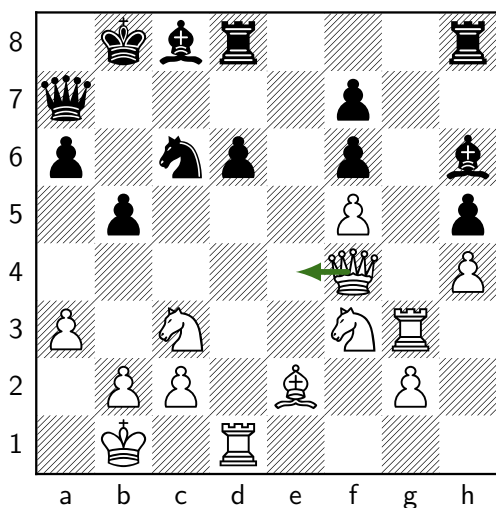
Figure 13: **Move 19, Black: Rb8.** (a) Our position; Melanie plays Rb8 (a8b8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3863` (ply 45) [structural distance $d=102$, QAP obj=73]: the master played Re8; under μ our *Ra1* plays the role of that game’s *Ra1*; cross-type roles: our *Bc8* as their *Pa6*, our *Re8* as their *Bd6*; transferred move a8b8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1885` (ply 26) [structural distance $d=102$, QAP obj=61]: the master played Rg1; under μ our *Ra1* plays the role of that game’s *Rh1*; cross-type roles: our *Pc7* as their *Rd1*, our *pf4* as their *Bf1*; transferred move a8b8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3021` (ply 30) [structural distance $d=104$, QAP obj=60]: the master played Rg1; under μ our *Ra1* plays the role of that game’s *Rh1*; cross-type roles: our *Ph6* as their *Bf1*, our *Bc8* as their *Pa2*; transferred move a8b8 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Rb8 (summed QAP objective 195, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



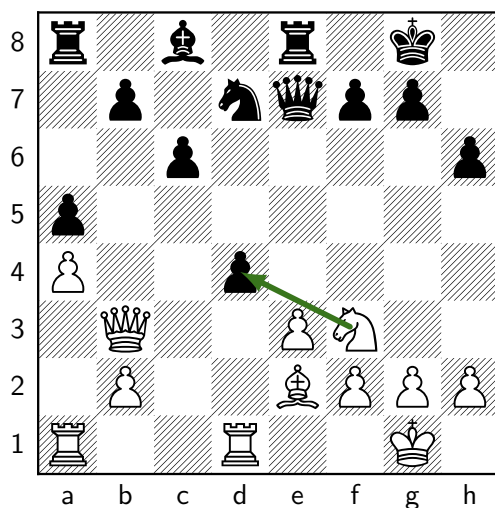
(a)



(b)

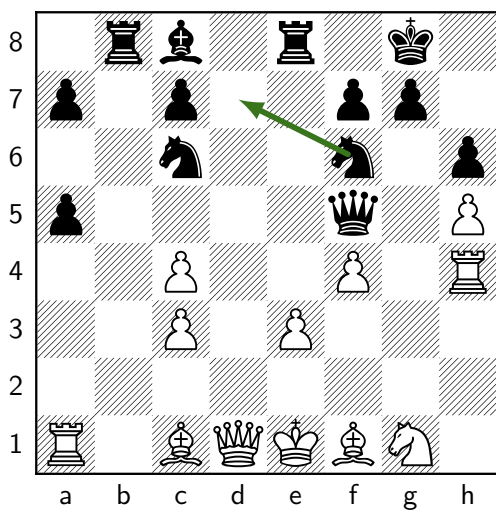


(c)

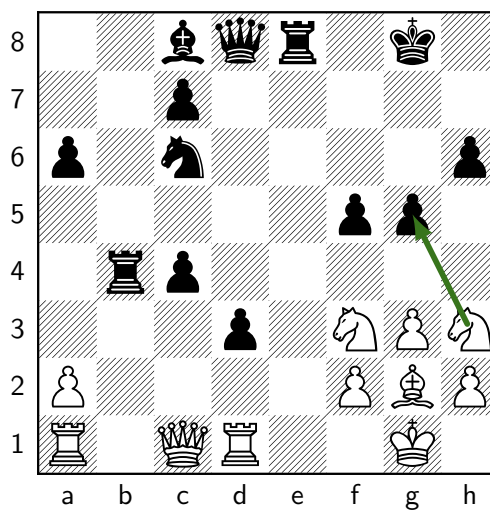


(d)

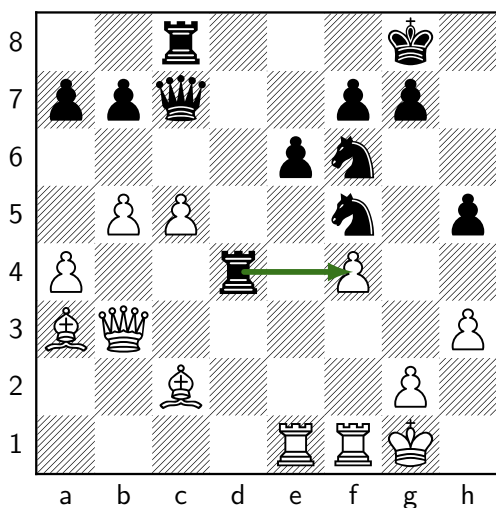
Figure 14: **Move 20, Black: Qf5.** (a) Our position; Melanie plays Qf5 (c5f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0002#673 (ply 46) [structural distance $d=112$, QAP obj=90]: the master played Nd4; under μ our Qc4 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our Ph6 as their Ra1, our Qc5 as their Bc1; transferred move c5f5 (**endorses this move**). (c) Retrieved from ccr1_top_p0003#1884 (ply 48) [structural distance $d=113$, QAP obj=66]: the master played Qe4; under μ our Qc4 plays the role of that game’s Qf4; cross-type roles: our Nf6 as their Be2, our Rb8 as their Pg2; transferred move c5f5 (**endorses this move**). (d) Retrieved from ccr1_top_p0001#876 (ply 32) [structural distance $d=109$, QAP obj=63]: the master played Nxd4; under μ our Qc4 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our Nf6 as their Ra1, our Qc5 as their Rd1; transferred move c5f5 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Qf5 (summed QAP objective 276, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



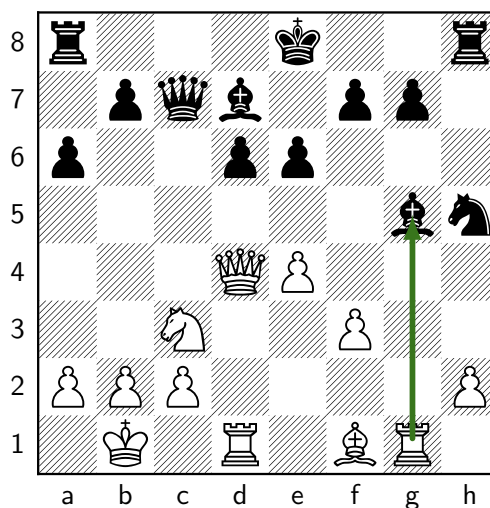
(a)



(b)

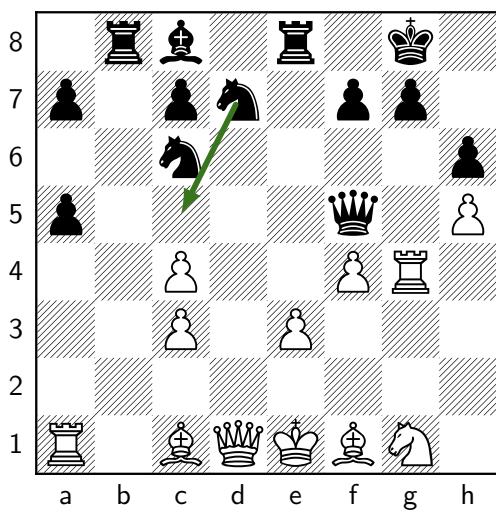


(c)

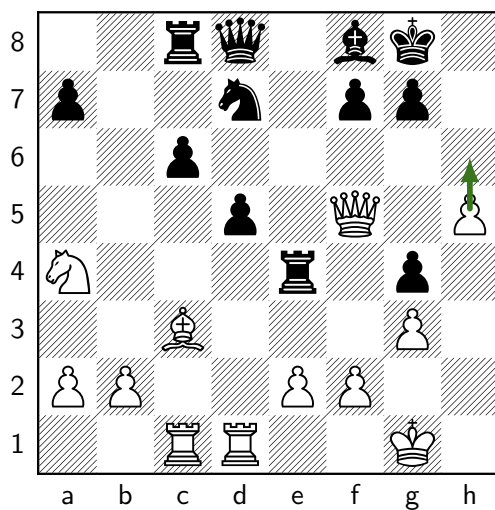


(d)

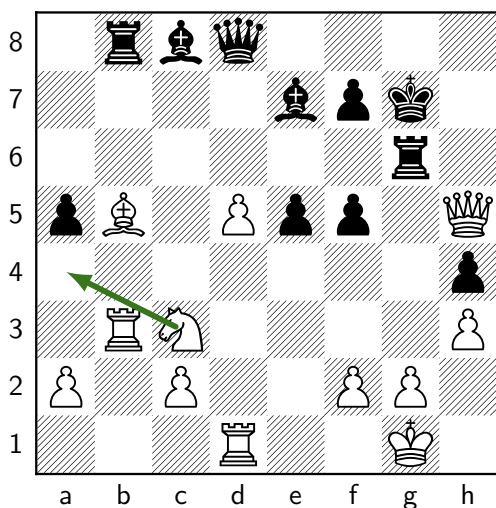
Figure 15: **Move 21, Black: Nd7**. (a) Our position; Melanie plays Nd7 (f6d7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2159` (ply 46) [structural distance $d=113$, QAP obj=69]: the master played Nhxg5; under μ our Nf3 plays the role of that game’s Nh3; cross-type roles: our Bc8 as their Ra1, our Qf5 as their Rd1; transferred move f6d7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3041` (ply 53) [structural distance $d=98$, QAP obj=63]: the master played Rxf4; under μ our Nf3 plays the role of that game’s Rd5 (a **cross-type** correspondence); cross-type roles: our Rb8 as their Pa7, our pf4 as their Nf5; transferred move f6d7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1885` (ply 28) [structural distance $d=91$, QAP obj=54]: the master played Rxxg5; under μ our Nf3 plays the role of that game’s Rg1 (a **cross-type** correspondence); cross-type roles: our Bc8 as their Rd1, our Nc6 as their Bf1; transferred move f6d7 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Nd7 (summed QAP objective 333, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



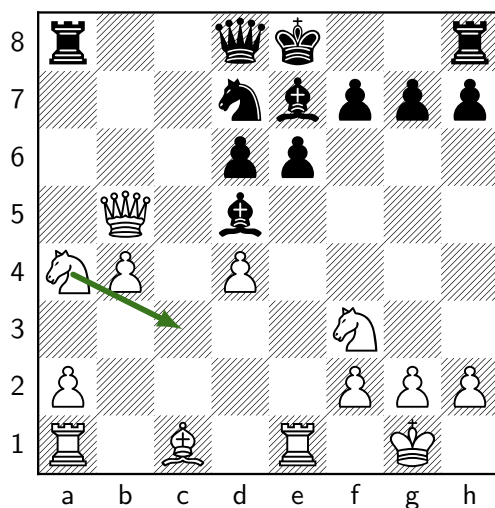
(a)



(b)

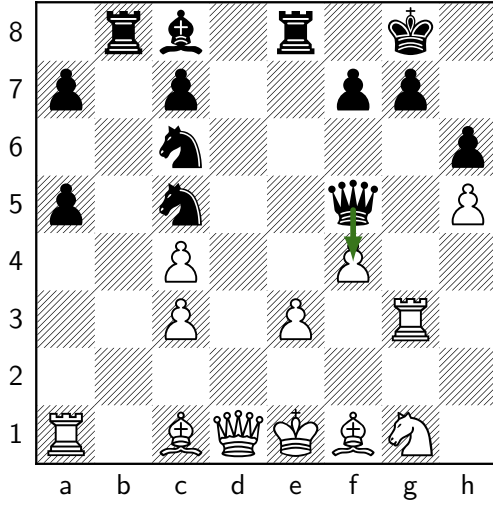


(c)

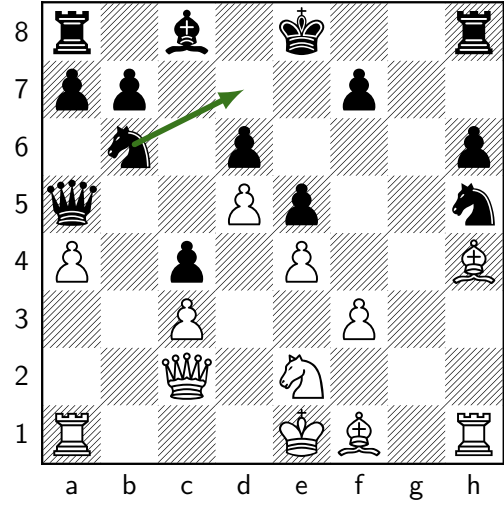


(d)

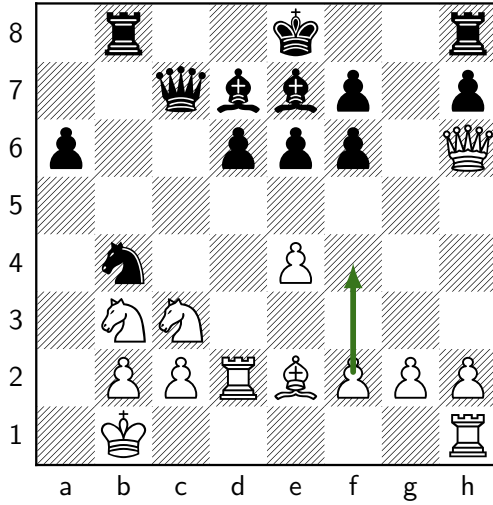
Figure 16: **Move 22, Black: Nc5**. (a) Our position; Melanie plays Nc5 (d7c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#3990` (ply 46) [structural distance $d=91$, QAP obj=84]: the master played h6; under μ our Nd2 plays the role of that game’s Ph5 (a **cross-type** correspondence); cross-type roles: our Pa5 as their Rc1, our bc1 as their Pb2; transferred move d7c5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#2811` (ply 50) [structural distance $d=100$, QAP obj=76]: the master played Na4; under μ our Nd2 plays the role of that game’s Nc3; cross-type roles: our Pa5 as their Rd1, our Rb8 as their Pc2; transferred move d7c5 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#399` (ply 34) [structural distance $d=100$, QAP obj=49]: the master played Nc3; under μ our Nd2 plays the role of that game’s Na4; cross-type roles: our pc3 as their Bc1, our Bc8 as their Ph2; transferred move d7c5 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Nc5 (summed QAP objective 223, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



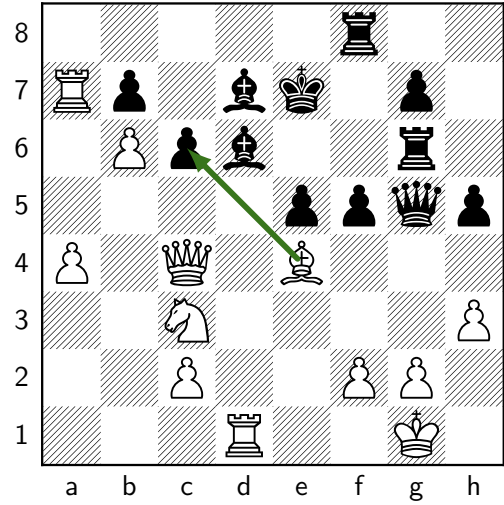
(a)



(b)

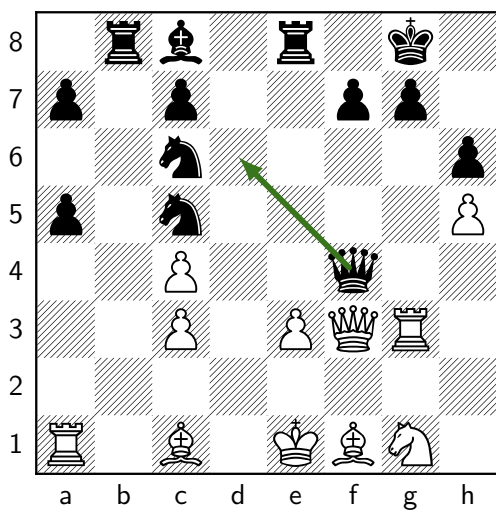


(c)

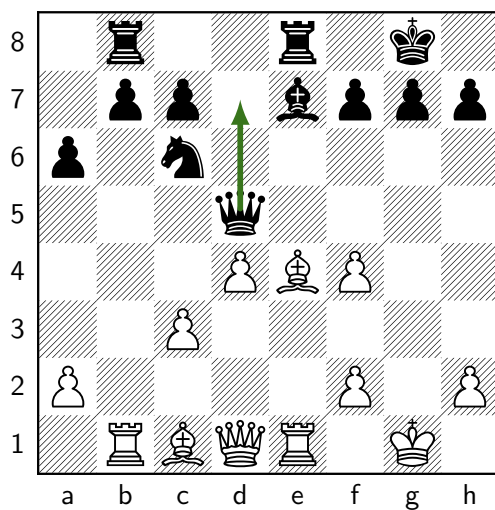


(d)

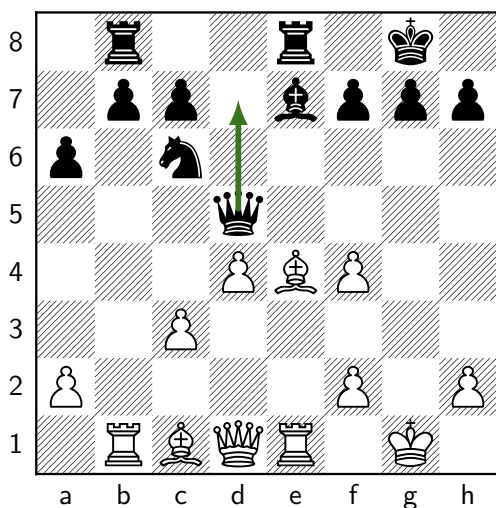
Figure 17: **Move 23, Black: Qxf4.** (a) Our position; Melanie plays Qxf4 (f5f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#553` (ply 37) [structural distance $d=90$, QAP obj=111]: the master played Nd7; under μ our Qf4 plays the role of that game’s Nb3 (a **cross-type** correspondence); cross-type roles: our Bc8 as their Ra8, our Nc5 as their Bc8; transferred move f5h7. (c) Retrieved from `ccrl_top_p0001#2161` (ply 32) [structural distance $d=107$, QAP obj=73]: the master played f4; under μ our Pa2 plays the role of that game’s Pf2; cross-type roles: our Nc6 as their Pb2, our Pf7 as their Be2; transferred move a7a6. (d) Retrieved from `ccrl_top_p0001#2390` (ply 52) [structural distance $d=104$, QAP obj=65]: the master played Bxc6; under μ our Qf4 plays the role of that game’s Be4 (a **cross-type** correspondence); cross-type roles: our Nc5 as their Pc2, our Pf7 as their Nc3; transferred move f5d5. Synthesis: no retrieved analog transferred to Qxf4 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



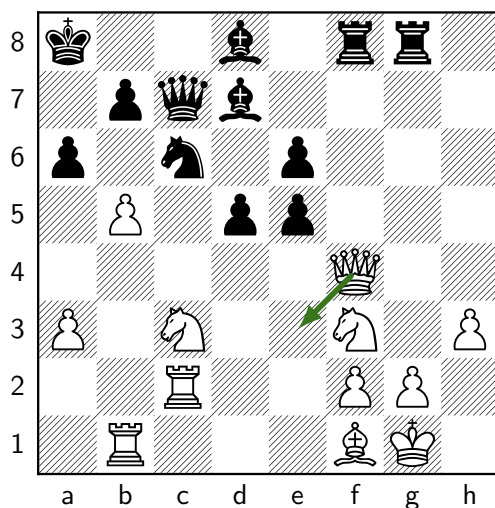
(a)



(b)

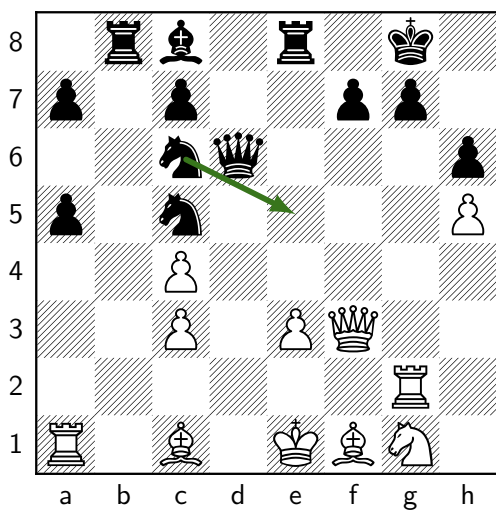


(c)

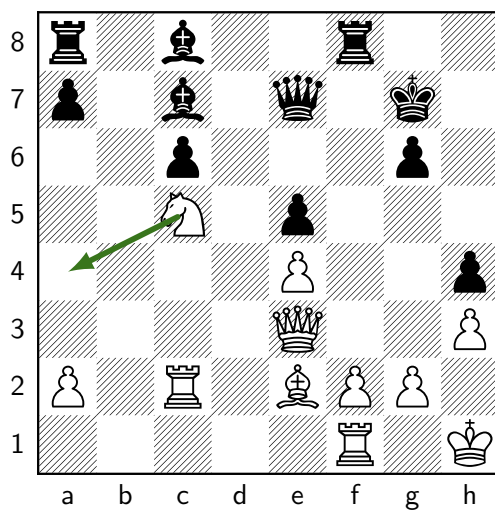


(d)

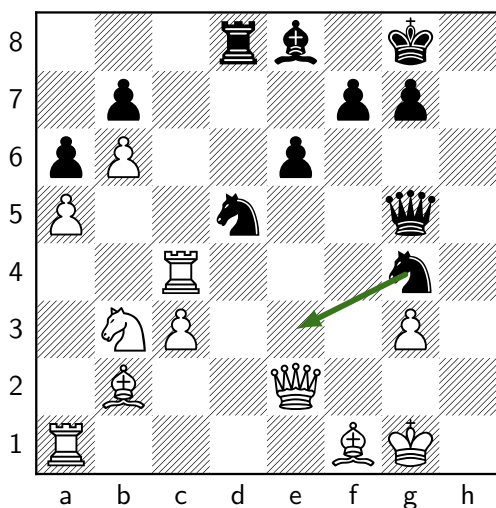
Figure 18: **Move 24, Black: Qd6.** (a) Our position; Melanie plays Qd6 (f4d6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2557` (ply 31) [structural distance $d=111$, QAP obj=88]: the master played Qd7; under μ our Qf5 plays the role of that game’s Qd4; cross-type roles: our Nc5 as their Ph7, our Re8 as their Nc6; transferred move f4d6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3891` (ply 31) [structural distance $d=111$, QAP obj=88]: the master played Qd7; under μ our Qf5 plays the role of that game’s Qd4; cross-type roles: our Nc5 as their Ph7, our Re8 as their Nc6; transferred move f4d6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2600` (ply 62) [structural distance $d=113$, QAP obj=71]: the master played Qe3; under μ our Qf5 plays the role of that game’s Qf4; cross-type roles: our Bc8 as their Rb1, our Nc6 as their Bf1; transferred move f4d6 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Qd6 (summed QAP objective 276, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



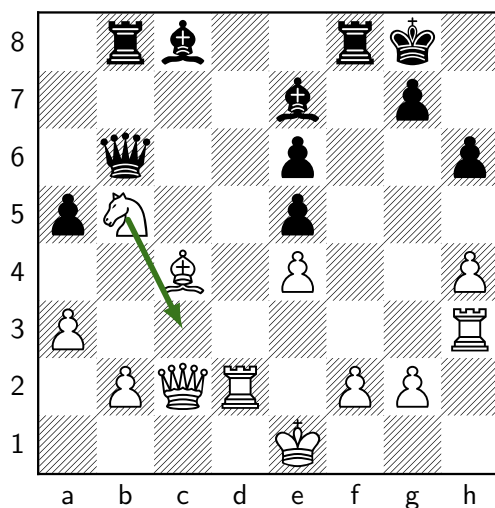
(a)



(b)

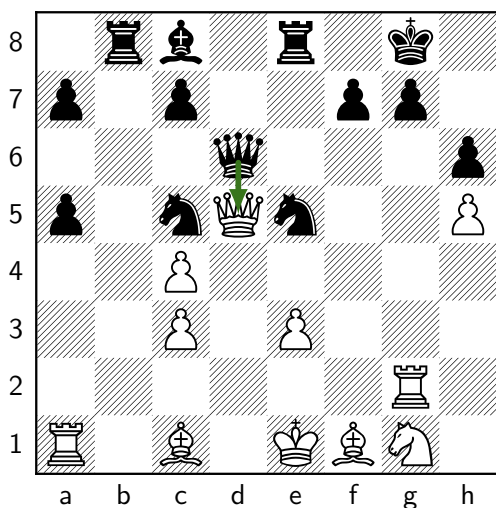


(c)

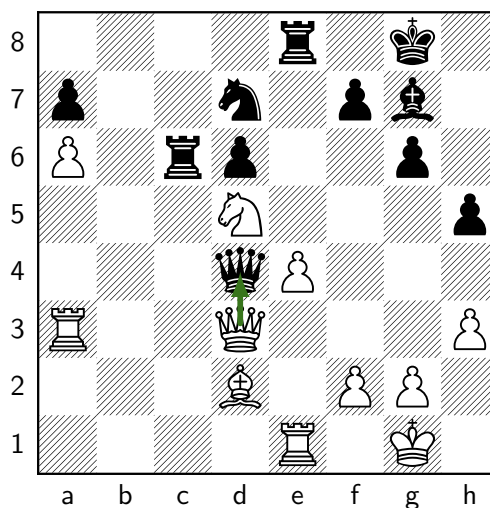


(d)

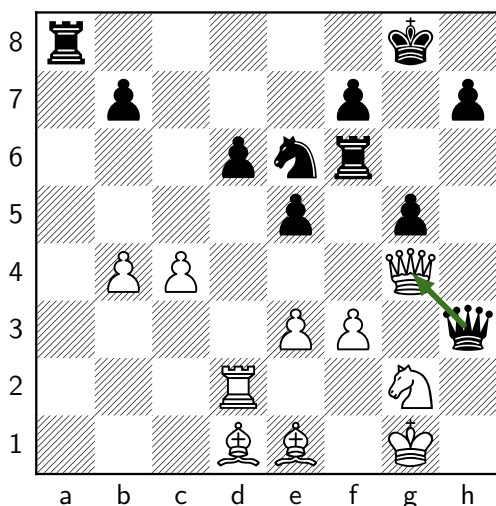
Figure 19: **Move 25, Black: Ne5**. (a) Our position; Melanie plays Ne5 (c6e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1928` (ply 58) [structural distance $d=88$, QAP obj=40]: the master played Na4; under μ our Nc3 plays the role of that game’s Nc5; cross-type roles: our Bc8 as their Pa2, our Pa7 as their Rc2; transferred move c6e5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2996` (ply 61) [structural distance $d=103$, QAP obj=31]: the master played Nge3; under μ our Nc3 plays the role of that game’s Ng5; cross-type roles: our Pf7 as their Rd8, our ng1 as their Be8; transferred move c6e5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1186` (ply 40) [structural distance $d=106$, QAP obj=23]: the master played Nc3; under μ our Nc3 plays the role of that game’s Nb5; cross-type roles: our ra1 as their Qc2, our Qd6 as their Rd2; transferred move c6e5 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Ne5 (summed QAP objective 95, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



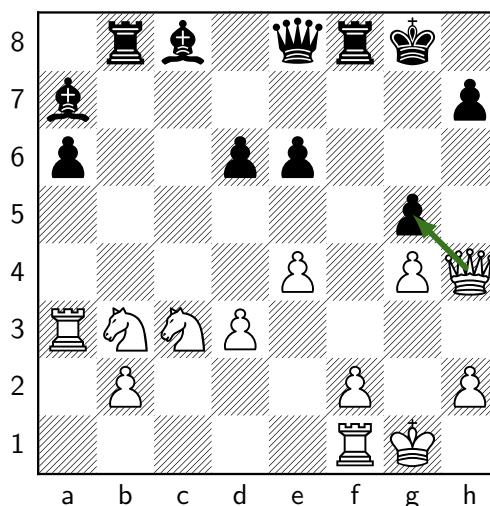
(a)



(b)

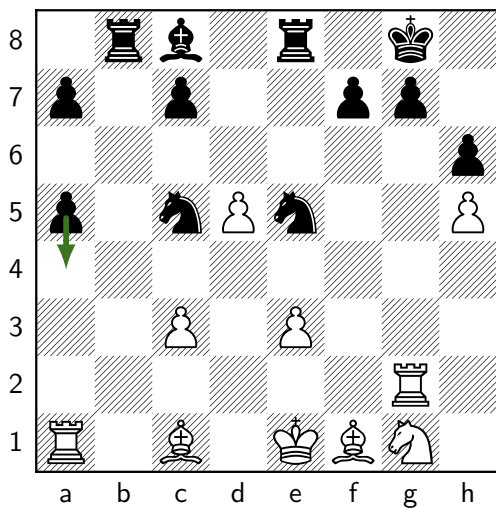


(c)

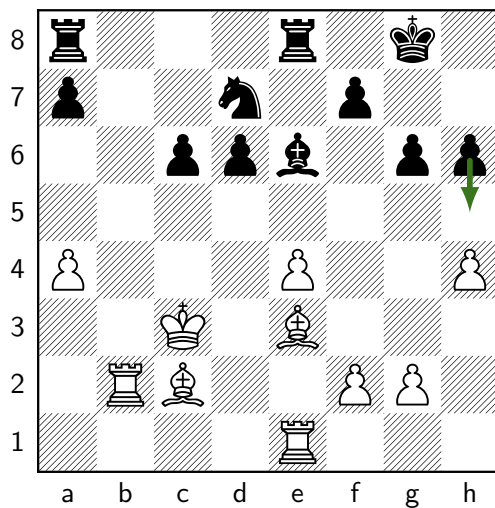


(d)

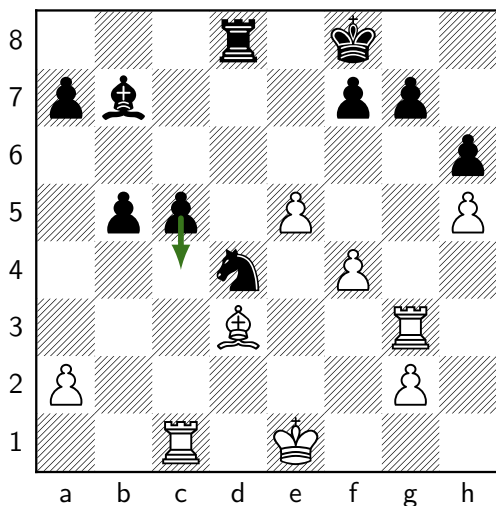
Figure 20: **Move 26, Black: Qxd5.** (a) Our position; Melanie plays Qxd5 (d6d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#549` (ply 62) [structural distance $d=101$, QAP obj=134]: the master played Qxd4; under μ our Qd3 plays the role of that game’s Qd3; cross-type roles: our Ph6 as their Ra3, our Ne5 as their Pe4; transferred move d6d5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1001` (ply 61) [structural distance $d=115$, QAP obj=103]: the master played Qxg4; under μ our Qd3 plays the role of that game’s Qh6; cross-type roles: our Rb8 as their Pb7, our ng1 as their Pf7; transferred move d6d5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2607` (ply 42) [structural distance $d=105$, QAP obj=44]: the master played Qxg5+; under μ our Qd3 plays the role of that game’s Qh4; cross-type roles: our Bc8 as their Ph2, our qd5 as their Ra3; transferred move d6d5 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Qxd5 (summed QAP objective 280, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



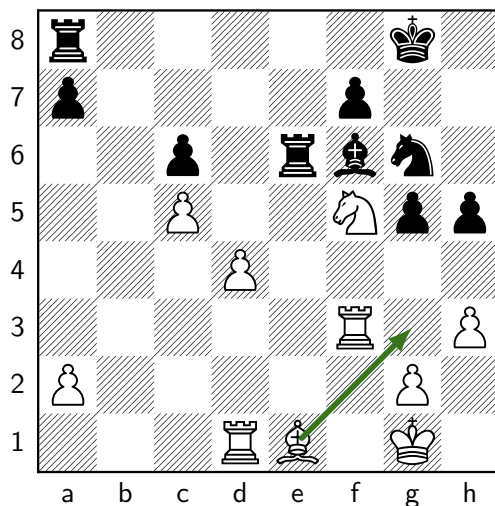
(a)



(b)

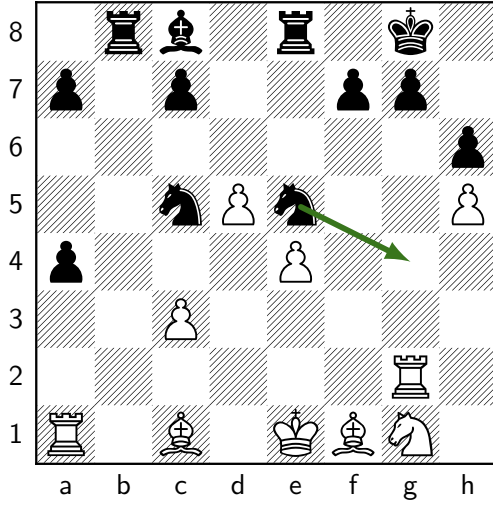


(c)

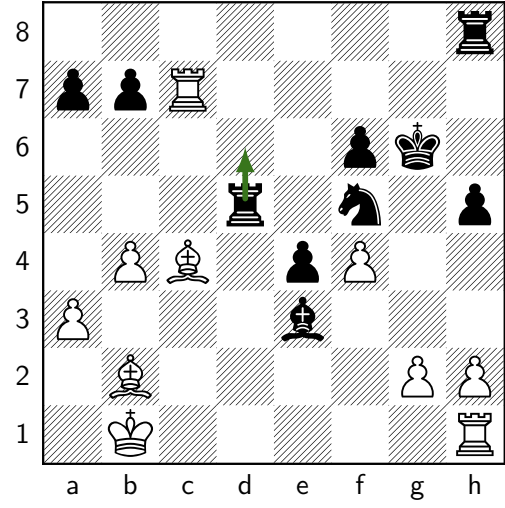


(d)

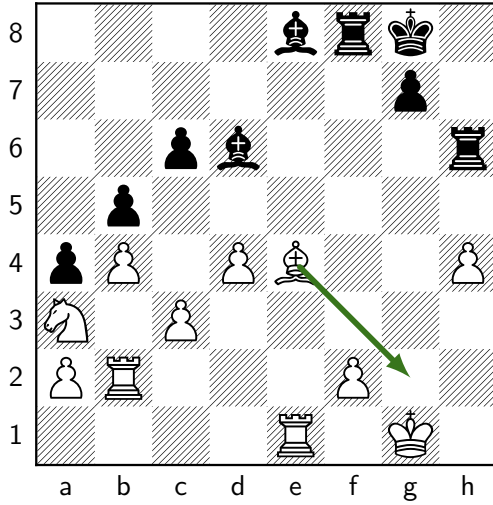
Figure 21: **Move 27, Black: a4**. (a) Our position; Melanie plays a4 (a5a4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#865` (ply 59) [structural distance $d=75$, QAP obj=31]: the master played h5; under μ our Pa_4 plays the role of that game’s Ph_3 ; cross-type roles: our Pa_7 as their Ra_8 , our $b1$ as their Nd_7 ; transferred move a5a4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#348` (ply 55) [structural distance $d=76$, QAP obj=29]: the master played c4; under μ our Pa_4 plays the role of that game’s Pc_4 ; cross-type roles: our Rb_8 as their Pa_7 , our Nc_5 as their Bb_7 ; transferred move a5a4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3632` (ply 56) [structural distance $d=77$, QAP obj=20]: the master played Bg3; under μ our Pa_4 plays the role of that game’s Be_1 (a **cross-type** correspondence); cross-type roles: our Pa_7 as their Rd_1 , our Rb_8 as their Pa_2 ; transferred move a5a4 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to a4 (summed QAP objective 93, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



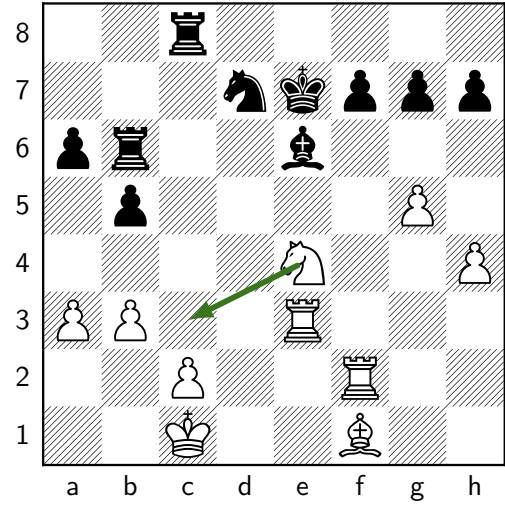
(a)



(b)

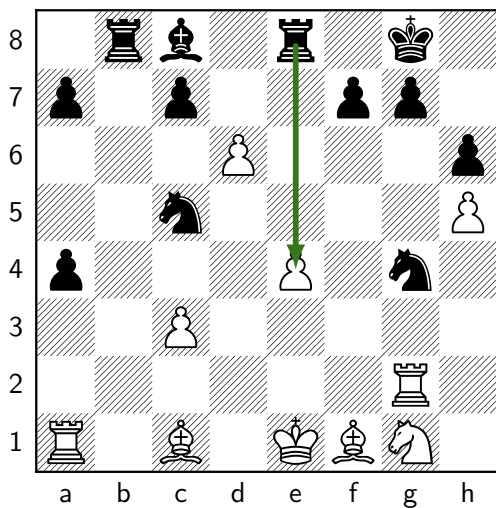


(c)

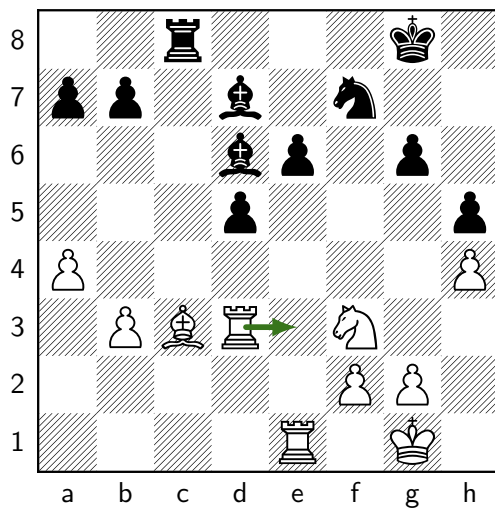


(d)

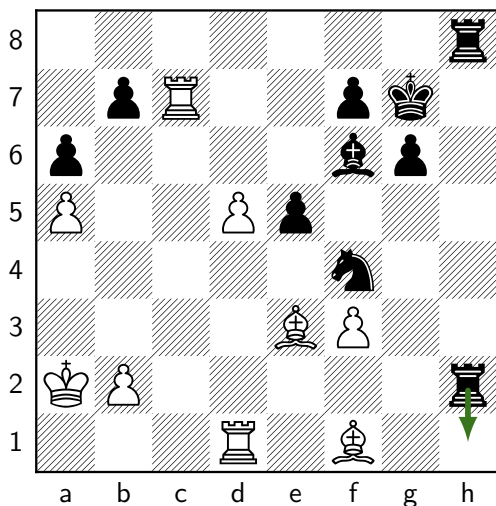
Figure 22: **Move 28, Black: Ng4.** (a) Our position; Melanie plays Ng4 (e5g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#76` (ply 49) [structural distance $d=70$, QAP obj=23]: the master played Rd6; under μ our *Ne4* plays the role of that game’s *Rd4* (a **cross-type** correspondence); cross-type roles: our *Pa7* as their *Rh8*, our *Re8* as their *Nf5*; transferred move e5g4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#315` (ply 60) [structural distance $d=67$, QAP obj=61]: the master played Bg2; under μ our *Nc4* plays the role of that game’s *Be4* (a **cross-type** correspondence); cross-type roles: our *Bc8* as their *Rb2*, our *Ne5* as their *Pc3*; transferred move c5b3. (d) Retrieved from `ccr1_top_p0003#1279` (ply 54) [structural distance $d=76$, QAP obj=60]: the master played Nc3; under μ our *Ne4* plays the role of that game’s *Ne4*; cross-type roles: our *Rb8* as their *Pa3*, our *Pf7* as their *Re3*; transferred move e5c6. Synthesis: 1 retrieved analog transferred to Ng4 (summed QAP objective 23, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



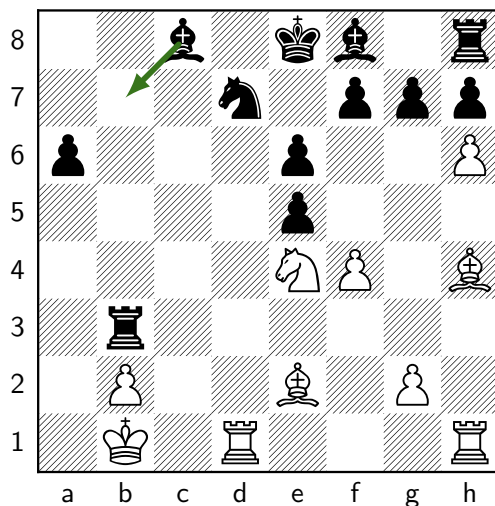
(a)



(b)

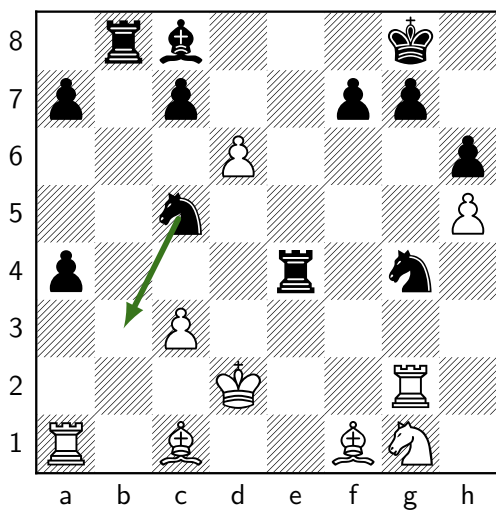


(c)

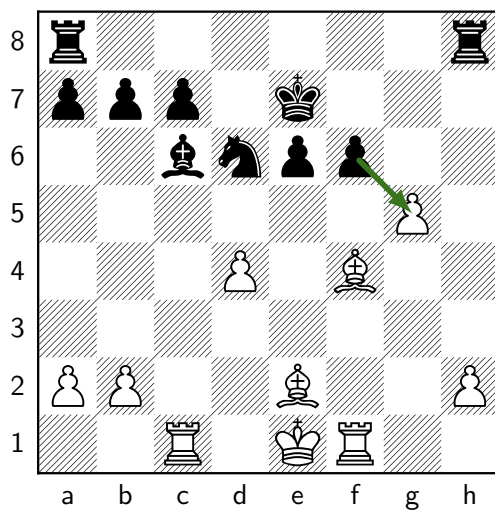


(d)

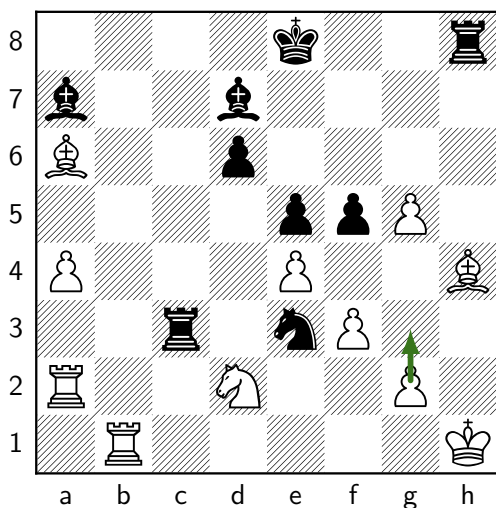
Figure 23: **Move 29, Black: Rxe4+.** (a) Our position; Melanie plays Rxe4+ (e8e4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#3625 (ply 62) [structural distance $d=79$, QAP obj=29]: the master played Rde3; under μ our Re1 plays the role of that game’s Rd3; cross-type roles: our Ph6 as their Re1, our Nc5 as their Pb3; transferred move e8e4 (**endorses this move**). (c) Retrieved from ccr1_top_p0003#1592 (ply 57) [structural distance $d=82$, QAP obj=29]: the master played Rh1; under μ our Re1 plays the role of that game’s Rh7; cross-type roles: our ng1 as their Rh8, our Pc7 as their rc7; transferred move e8e4 (**endorses this move**). (d) Retrieved from ccr1_top_p0000#2294 (ply 43) [structural distance $d=68$, QAP obj=55]: the master played Bb7; under μ our Pa2 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our Ph6 as their Bf8, our Bc8 as their Rh8; transferred move a7a6. Synthesis: 2 retrieved analogs transferred to Rxe4+ (summed QAP objective 58, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



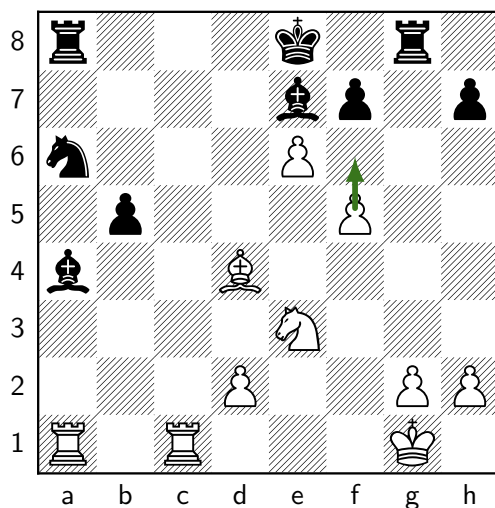
(a)



(b)



(c)



(d)

Figure 24: **Move 30, Black: Nb3+.** (a) Our position; Melanie plays Nb3+ (c5b3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#46` (ply 57) [structural distance $d=78$, QAP obj=45]: the master played **fxg5**; under μ our *Pc2* plays the role of that game’s *Pf3*; cross-type roles: our *Pg7* as their *Rh8*, our *ra1* as their *Bc6*; transferred move **c7d6**. (c) Retrieved from `ccr1_top_p0002#25` (ply 56) [structural distance $d=72$, QAP obj=42]: the master played **g3**; under μ our *Pa5* plays the role of that game’s *Pg2*; cross-type roles: our *Pg7* as their *Nd2*, our *Bc8* as their *rc3*; transferred move **a4a3**. (d) Retrieved from `ccr1_top_p0002#2200` (ply 56) [structural distance $d=56$, QAP obj=41]: the master played **f6**; under μ our *Nc4* plays the role of that game’s *Pf5* (a **cross-type** correspondence); cross-type roles: our *ng1* as their *Rc1*, our *Re4* as their *Ph2*; transferred move **c5a6**. Synthesis: no retrieved analog transferred to Nb3+ – the **search selector** chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.

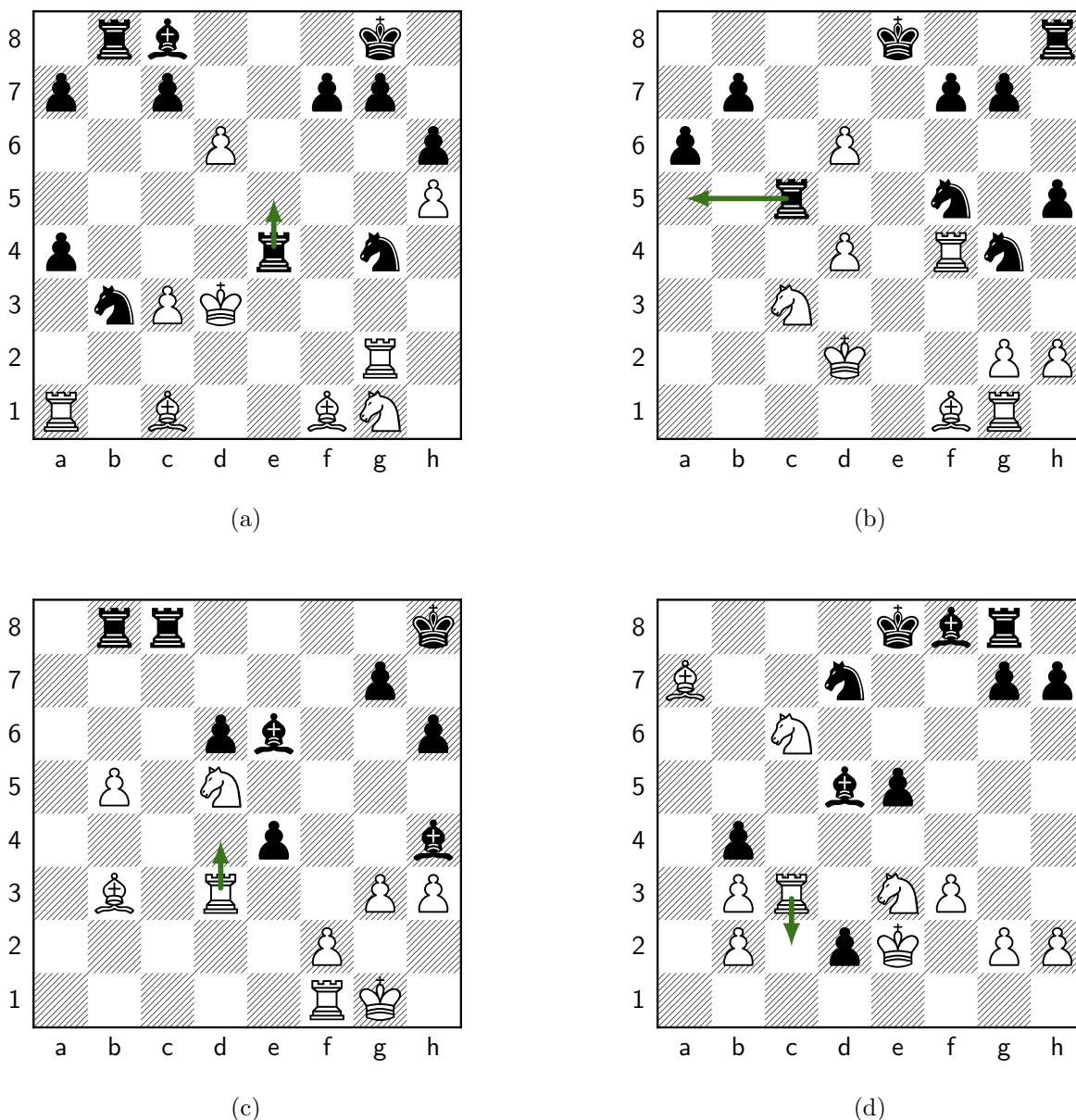
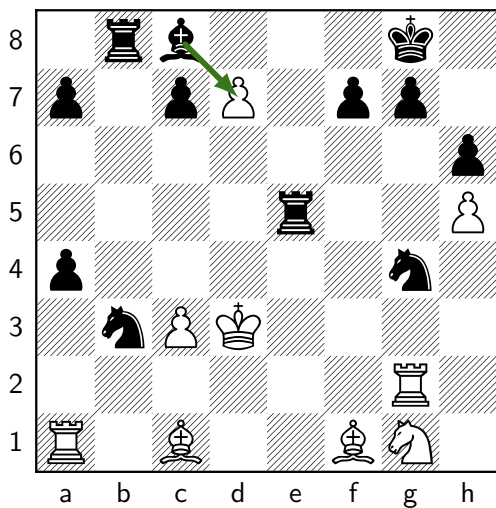
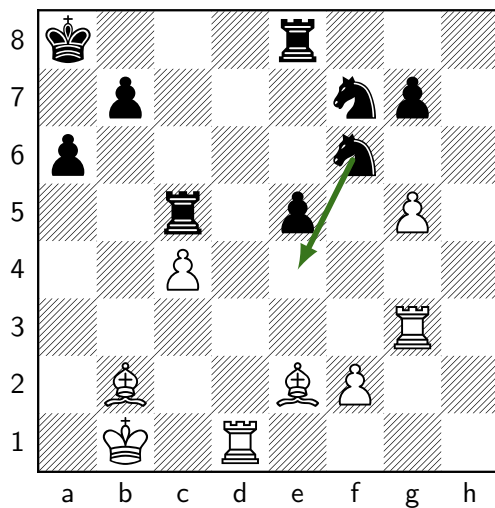


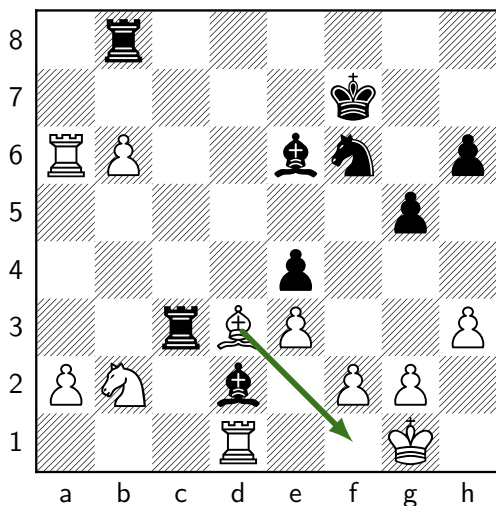
Figure 25: **Move 31, Black: Re5.** (a) Our position; Melanie plays Re5 (e4e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#748` (ply 57) [structural distance $d=84$, QAP obj=37]: the master played Ra5; under μ our Re5 plays the role of that game’s Rc4; cross-type roles: our Bc8 as their Rh8, our bfl as their Pf7; transferred move e4e5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1977` (ply 66) [structural distance $d=82$, QAP obj=33]: the master played Rd4; under μ our Re5 plays the role of that game’s Rd3; cross-type roles: our Pf7 as their Rf1, our Nb3 as their Pg3; transferred move e4e5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#513` (ply 56) [structural distance $d=89$, QAP obj=40]: the master played Rc2; under μ our Re5 plays the role of that game’s Rc3; cross-type roles: our Bc8 as their Pg2, our Rb8 as their Ph2; transferred move e4e8. Synthesis: 2 retrieved analogs transferred to Re5 (summed QAP objective 70, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



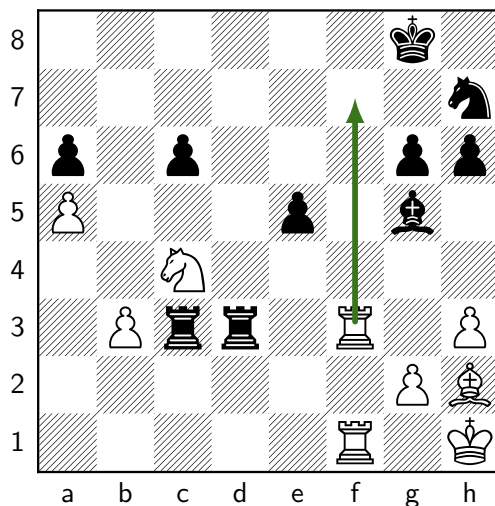
(a)



(b)

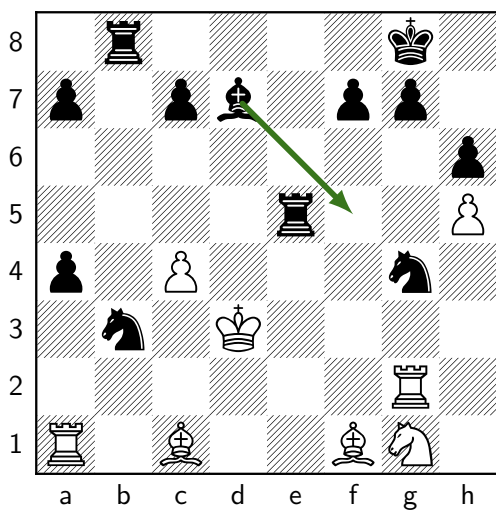


(c)

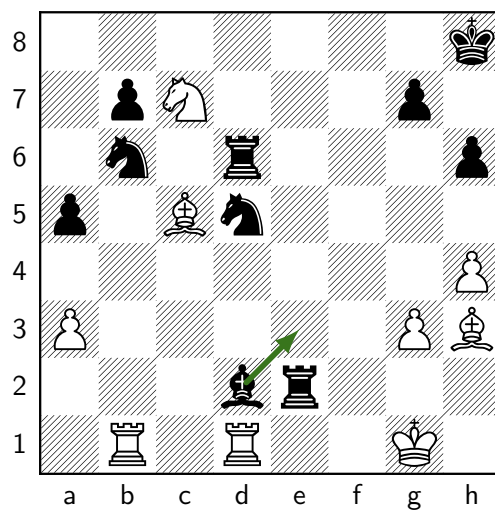


(d)

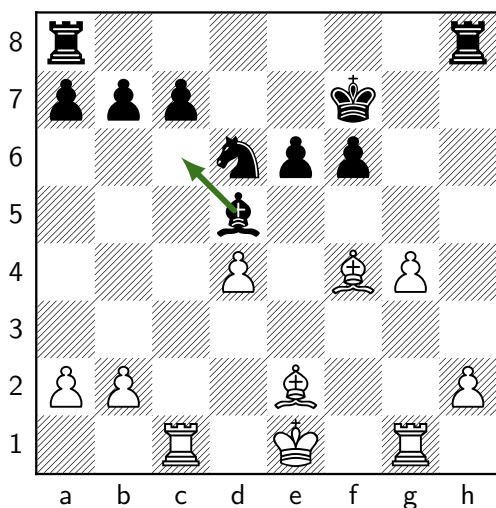
Figure 26: **Move 32, Black: Bxd7.** (a) Our position; Melanie plays Bxd7 (c8d7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#91` (ply 79) [structural distance $d=75$, QAP obj=29]: the master played Nf4; under μ our Bc1 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our Rb8 as their Pg7, our bc1 as their Pa6; transferred move c8d7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#673` (ply 64) [structural distance $d=77$, QAP obj=57]: the master played Bf1; under μ our Bc1 plays the role of that game’s Bd3; cross-type roles: our Nb3 as their Rd1, our Rb8 as their Nb2; transferred move c8b7. (d) Retrieved from `ccr1_top_p0002#2266` (ply 74) [structural distance $d=70$, QAP obj=50]: the master played Rf7; under μ our Bc1 plays the role of that game’s Rf3 (a **cross-type** correspondence); cross-type roles: our ph5 as their Bh2, our bc1 as their Pb3; transferred move c8a6. Synthesis: 1 retrieved analog transferred to Bxd7 (summed QAP objective 29, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



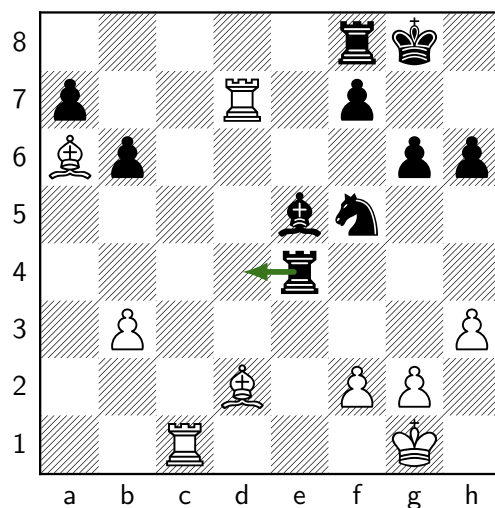
(a)



(b)

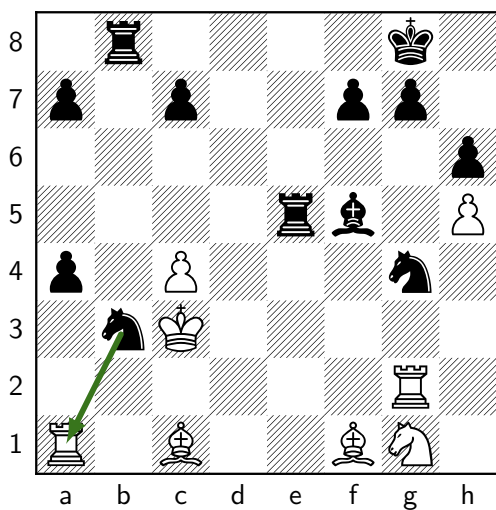


(c)

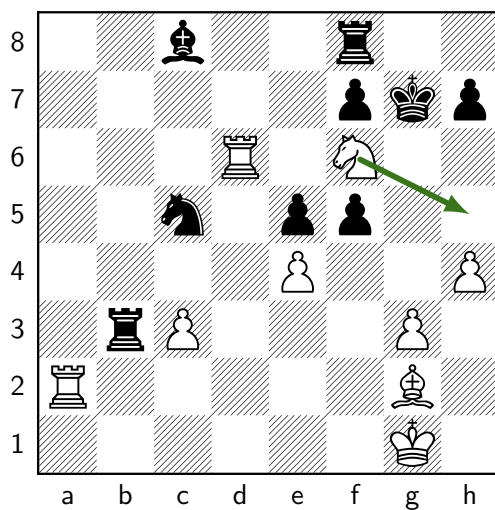


(d)

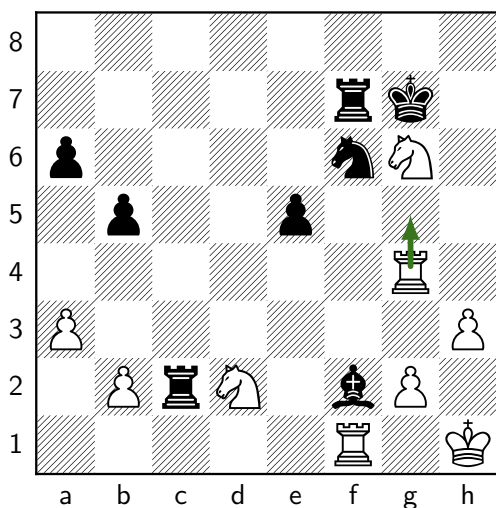
Figure 27: **Move 33, Black: Bf5+.** (a) Our position; Melanie plays Bf5+ (d7f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#3199` (ply 63) [structural distance $d=73$, QAP obj=31]: the master played Be3+; under μ our Bd2 plays the role of that game’s Bd7; cross-type roles: our bc1 as their Nb6, our Nb3 as their bc5; transferred move d7f5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#46` (ply 53) [structural distance $d=73$, QAP obj=15]: the master played Bc6; under μ our Bd2 plays the role of that game’s Bd4; cross-type roles: our ra1 as their Nd6, our Ng4 as their Pf6; transferred move d7f5 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#1620` (ply 57) [structural distance $d=73$, QAP obj=4]: the master played Rd4; under μ our Bd2 plays the role of that game’s Re5 (a **cross-type** correspondence); cross-type roles: our Nb3 as their rd7, our Rb8 as their ba6; transferred move d7f5 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Bf5+ (summed QAP objective 55, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



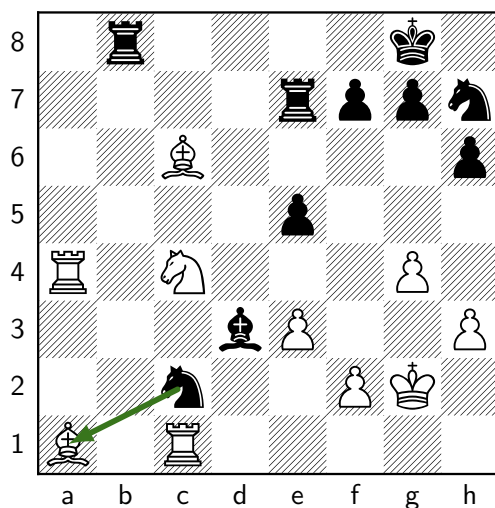
(a)



(b)

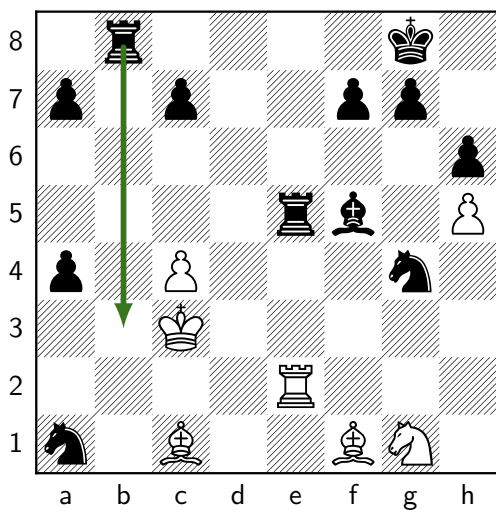


(c)

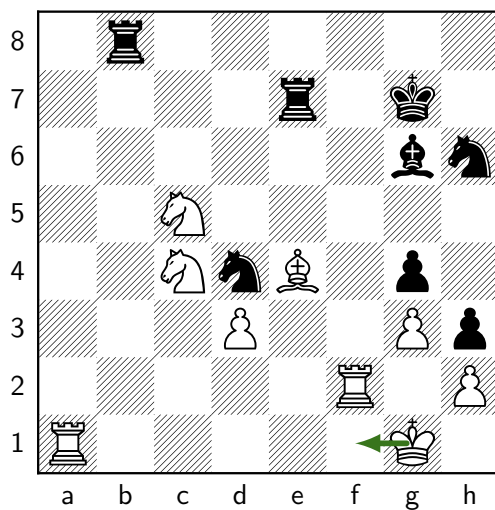


(d)

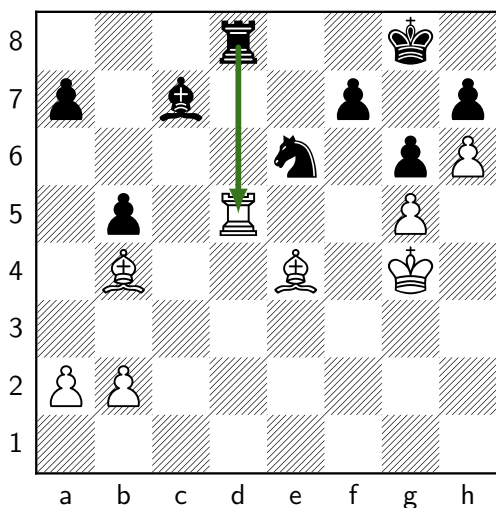
Figure 28: **Move 34, Black: Nxa1.** (a) Our position; Melanie plays Nxa1 (b3a1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#12` (ply 64) [structural distance $d=77$, QAP obj=37]: the master played Nh5+; under μ our Nb6 plays the role of that game’s Nf6; cross-type roles: our Pa7 as their Ra2, our Nb3 as their rb3; transferred move b3a1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#754` (ply 64) [structural distance $d=77$, QAP obj=57]: the master played Rg5; under μ our Ng5 plays the role of that game’s Rg4 (a **cross-type** correspondence); cross-type roles: our kc3 as their Rf1, our bc1 as their Pb2; transferred move g4f2. (d) Retrieved from `ccr1_top_p0000#658` (ply 73) [structural distance $d=76$, QAP obj=47]: the master played Nxa1; under μ our Nb6 plays the role of that game’s Nc7; cross-type roles: our Ng4 as their Pg7, our Pc7 as their Nh7; transferred move b3c1. Synthesis: 2 retrieved analogs transferred to Nxa1 (summed QAP objective 74, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



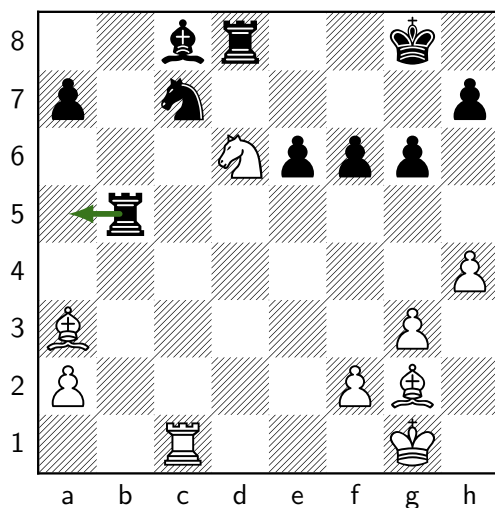
(a)



(b)

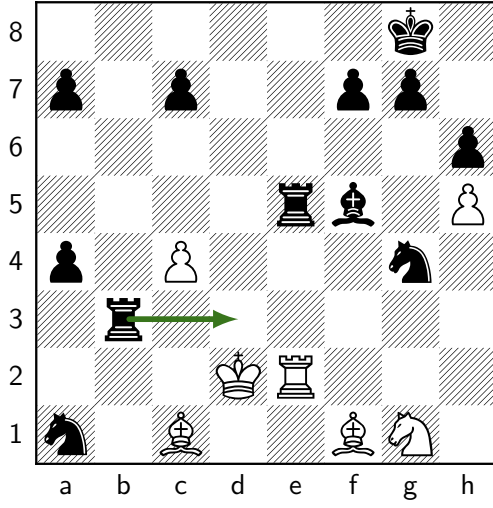


(c)

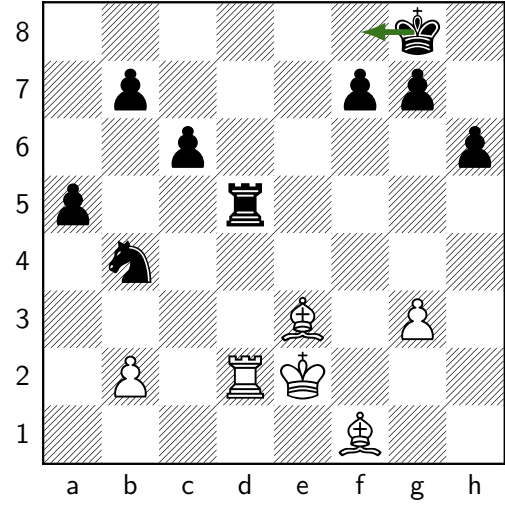


(d)

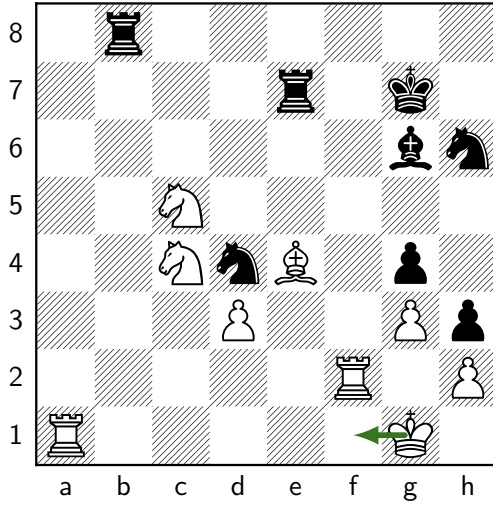
Figure 29: **Move 35, Black: Rb3+.** (a) Our position; Melanie plays Rb3+ (b8b3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#119` (ply 80) [structural distance $d=72$, QAP obj=46]: the master played Kf1; under μ our Kg1 plays the role of that game’s Kg1; cross-type roles: our Na1 as their Rf2, our Bf5 as their Pd3; transferred move g8f8. (c) Retrieved from `ccr1_top_p0002#3179` (ply 73) [structural distance $d=66$, QAP obj=43]: the master played Rxd5; under μ our Re4 plays the role of that game’s Rd1; cross-type roles: our bc1 as their Pf7, our Na1 as their Pg6; transferred move e5e2. (d) Retrieved from `ccr1_top_p0001#2961` (ply 59) [structural distance $d=70$, QAP obj=32]: the master played Ra5; under μ our Re4 plays the role of that game’s Rb4; cross-type roles: our Re5 as their Bc8, our re2 as their nd6; transferred move e5a5. Synthesis: no retrieved analog transferred to Rb3+ – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



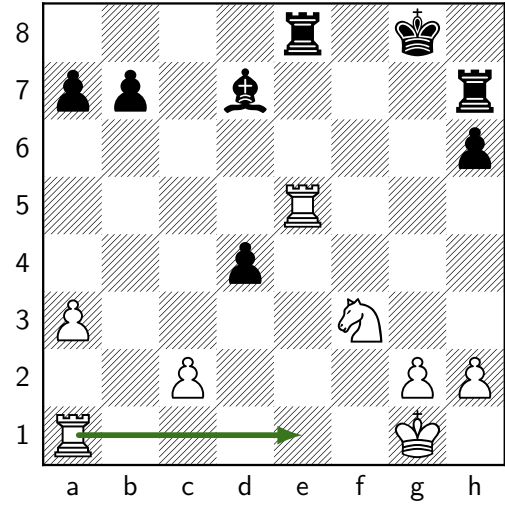
(a)



(b)

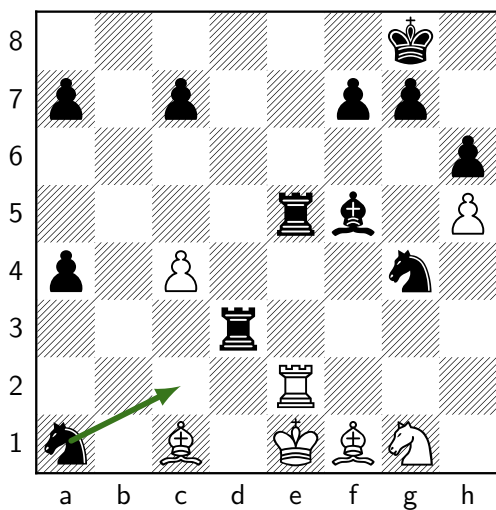


(c)

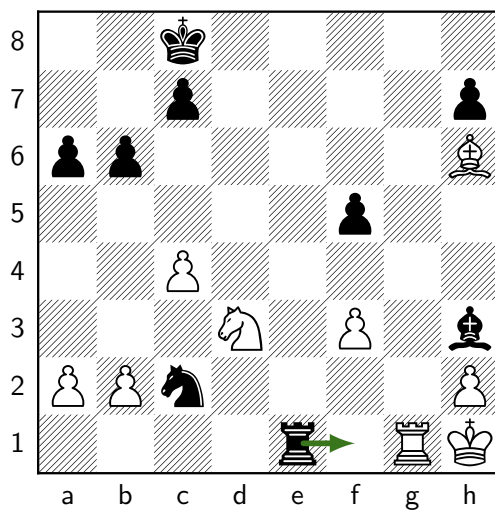


(d)

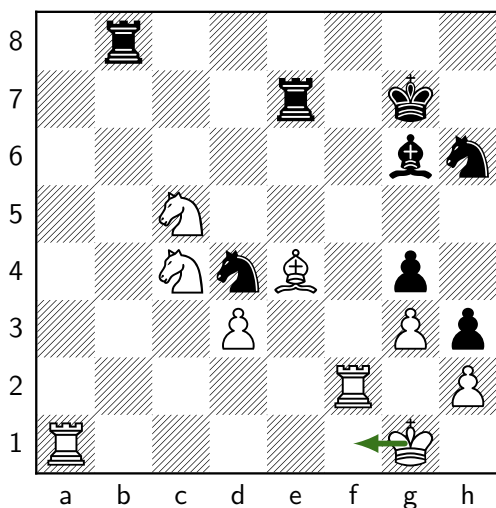
Figure 30: **Move 36, Black: Rd3+.** (a) Our position; Melanie plays Rd3+ (b3d3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#1176` (ply 81) [structural distance $d=102$, QAP obj=53]: the master played Kf8; under μ our Kg1 plays the role of that game’s Kg1; cross-type roles: our ng1 as their Pc6, our Rb3 as their Ph6; transferred move g8f8. (c) Retrieved from `ccrl_top_p0003#119` (ply 80) [structural distance $d=95$, QAP obj=49]: the master played Kf1; under μ our Kg1 plays the role of that game’s Kg1; cross-type roles: our Pa7 as their Ra1, our Bf5 as their Pd3; transferred move g8h7. (d) Retrieved from `ccrl_top_p0003#1758` (ply 56) [structural distance $d=101$, QAP obj=45]: the master played Rae1; under μ our Pg2 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Rb3 as their Pc2, our Bf5 as their Pa3; transferred move g7g6. Synthesis: no retrieved analog transferred to Rd3+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



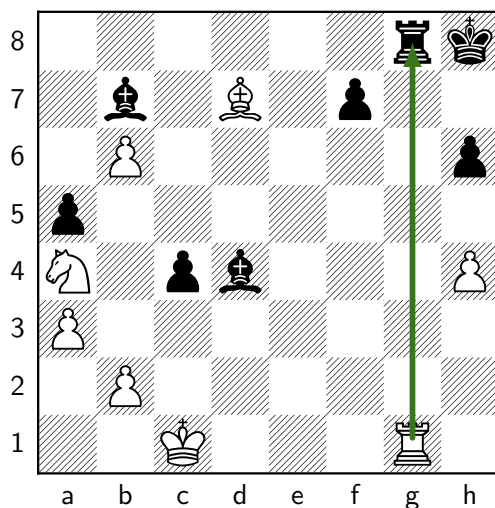
(a)



(b)



(c)



(d)

Figure 31: **Move 37, Black: Nc2#.** (a) Our position; Melanie plays Nc2# (a1c2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3007` (ply 59) [structural distance $d=79$, QAP obj=60]: the master played Rf1; under μ our *Re4* plays the role of that game’s *Re8*; cross-type roles: our *Na1* as their *Ph7*, our *Pc7* as their *bh6*; transferred move e5e4. (c) Retrieved from `ccr1_top_p0003#119` (ply 80) [structural distance $d=85$, QAP obj=48]: the master played Kf1; under μ our *Kg1* plays the role of that game’s *Kg1*; cross-type roles: our *Bf5* as their *Ra1*, our *bc1* as their *Rf2*; transferred move g8h7. (d) Retrieved from `ccr1_top_p0001#717` (ply 60) [structural distance $d=83$, QAP obj=41]: the master played Rxg8+; under μ our *Re4* plays the role of that game’s *Rg1*; cross-type roles: our *Re5* as their *Pb2*, our *ng1* as their *pc4*; transferred move e5e2. Synthesis: no retrieved analog transferred to Nc2# – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.